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Smart Logistic Services System

نظام خدمات لوجستية ذكي

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الآية

قال تعالى:-

﴿يَرْفَعُ اللَّهُ الَّذِينَ آمَنُوا مِنْكُمْ وَالَّذِينَ أُوتُوا الْعِلْمَ دَرَجَاتٍ وَاللَّهُ بِمَا تَعْمَلُونَ خَبِيرٌ﴾
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سورة المجادلة: الآية 11

الإهداء

اللهم صل و سلم و بارك على سيدنا و نبينا محمد و على آله و أصحابه
الطيبين الطاهرين

إلى صاحب السيرة العطرة، والفكر المُستنير؛ فلقد كان له الفضل الأول في
بلوغي التعليم العالي (والدي الحبيب)، أطال الله في عُمره.

إلى من وضعتني على طريق الحياة، وجعلتني رابط الجأش، وراعتني حتى
صرت كبيراً (أمي الغالية)، حفظها الله و رعاها.

إلى إخوتي؛ من كان لهم بالغ الأثر في كثير من العقبات والصعاب.

إلى جميع أساتذتي الكرام؛ ممن لم يتوانوا في مد يد العون لي أهدي إليكم
بحثي.

الشـكر و الـعـرفـان

نحمد الله حمداً كثيراً طيباً مباركاً فيه أن وفقنا في إتمام هذا البحث و أمدنا بالصحة و العافية و العزيمة و الإصرار.

نتقدم بجزيل الشكر و العرفان و التقدير إلى :-

الدكتورة المشـرفة "عفاف مدني" على تفانيها و كل ما قدمته لنا من توجيهات و معلومات قيمة ساهمت في إثراء موضوع بحثنا في جوانبه المختلفة جزاها الله عنا كل الخير.

كما نتوجه بالشكر و التقدير إلى أعضاء لجنة المناقشة الموقرة و كل من ساهم في مساعدتنا على إتمام هذا البحث.

Abstract

The logistics sector, which consists of shipping and transporting products and goods from the place of production to the place of consumption, is one of the most important sectors in the shipping and transport system, which has become as important as producing the product itself.

With the development of computer technologies and information technology and their entry into various fields such as industry and management and other fields and the emergence of service management software, it has become necessary to use software systems to manage logistics operations.

In Sudan, logistics services suffer from some problems in the process of distributing goods and products, as it is done in a traditional way. The traditional distribution depends on the mental preservation of the distribution areas or description, which leads to problems that make the distribution process more difficult and costly, in addition to the lack of control over the distribution process.

In this research, a software system was developed to manage and control the processes of ordering and distributing products along the distribution lines.

The overall goal of the project is to develop logistics services so that the process of transporting goods or products from the place of production to the consumer is done in a smart and smooth manner using digital maps. Thus, the time, effort and cost required for both the producer and the consumer are saved.

المستخلص

يعد قطاع الخدمات اللوجستية والذي هو عبارة عن عمليات شحن و نقل المنتجات و البضائع من مكان الإنتاج الى مكان الإستهلاك ،من أهم القطاعات في منظومة الشحن والنقل، والتي أصبحت أهميتها بنفس أهمية انتاج المنتج نفسه .

ومع تطور تقنيات الحاسوب وتقانة المعلومات ودخولها في شتى المجالات كالصناعة والإدارة وغيرها من المجالات وظهور برمجيات إدارة الخدمات، أصبح من الضروري استخدام أنظمة برمجية لإدارة عمليات الخدمات اللوجستية.

في السودان تعاني الخدمات اللوجستية من بعض المشاكل في عملية توزيع السلع والمنتجات حيث تتم بطريقة تقليدية،التوزيع التقليدي يعتمد على الحفظ الذهني لمناطق التوزيع أو الوصف مما يؤدي إلى حدوث مشاكل تجعل عملية التوزيع أكثر صعوبة و تكلفة إضافة إلى قلة الرقابة على عملية التوزيع. في هذا البحث تم تطوير نظام برمجي لإدارة و مراقبة عمليات طلب و توزيع المنتجات على طول خطوط التوزيع .

الهدف العام للمشروع هو تطوير الخدمات اللوجستية بحيث تتم عملية نقل البضائع أو المنتجات من مكان ا لإنتاج إلى المستهلك بطريقة ذكية وسلسة باستخدام الخرائط الرقمية. وبالتالي ، يتم توفير الوقت والجهد و التكلفة المطلوبة لكل من المنتج والمستهلك.

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CHAPTER ONE

INTRODUCTION

1.1 Overview

This chapter provides an introduction to the logistics services, and explains the scope of our project, the methodology used, the objectives and the importance of the project and expected outcomes.

1.2 Definition

Logistics services are all the elements of your supply chain, from the factory to the end customer. They include transportation from manufacturer to warehouse, warehousing and order fulfillment, and delivery to the end customer.

Logistics services include:

- Transport from the factory to the fulfillment warehouse.
- Warehousing and order fulfillment.
- Outbound shipping and order delivery to customers [1].

This document specifies the specifications and requirements regarding the digital map of the distribution centers of the producing companies located in the state of Khartoum that may adopt the system in the future.

The purpose is to provide clarity about the location of the distribution centers and the routes of vehicles, as opposed to the requirements of a traditional distribution.

1.3 Problem

It is noticeable that the process of distributing goods and products in Sudan is done in a traditional way. The traditional distribution depends on the mental preservation of the areas of distribution or description, which leads to the occurrence of problems that hinder the distribution process.

Mental preservation of the location of the product or description is impractical and prone to errors (or forgetfulness). This may lead to a delay in the arrival of the product in its specified time, which leads to consumer dissatisfaction and thus loss of customer confidence.

When adopting the traditional distribution methods and appointing a new

(distribution) staff, it is very difficult to complete the distribution process according to the schedule and available resources due to the lack of geographical knowledge of the distribution areas which is required by the traditional distribution. This increases the distribution budget, and consequently increases the price of the product, and may result in less or no demand for the product.

The absence of a smart distribution system that results in inefficient or almost non-existent control at some times, and products or commodities may be distributed in places where there is little or no demand for them.

1.4 Objectives

Development of a smart logistic services system through which the process of distributing products and goods is done in an optimal way and imposing control over the distribution process.

Reducing the cost and time required in the distribution process and optimizing the resources of the distribution companies.

This system provides the feature of adding new distribution (consumption) areas and approving them.

1.5 Importance of Project

Logistics services today are part of the supply chain management that ensures that companies are provided with the resources they need to manufacture products and distribute to markets and consumption points.

The presence of a smart distribution system helps a competitive marketing environment with a high degree of control, as well as helps cover large geographical areas for the distribution of products, and also makes the company able to take administrative marketing decisions that in turn contribute to increasing profits, as well as helping to reduce the cost of distributing the product or goods, the optimal utilization of resources, the optimal use of technology and the creation of smart digital maps.

1.6 Scope

The boundary of this project covers the process of ordering products by the distributors and the process of monitoring the distribution process which consists of transporting the products from the producing companies

until reaching the consumption places.

It doesn't cover the process of billing or the process of collecting returns of products and their belongings neither reverse logistics operations.

1.7 Expected Outcomes

Finding a solution to the problem of product distribution helps the producing companies to:

- o produce the largest possible amount of products and distribute them in the best way, thus increasing the profits of the companies and creating a competitive advantage for the companies providing the distribution service, and gaining the confidence and satisfaction of customers.
- o Knowing the areas in which there is a high demand for a particular product and increasing the rate of its distribution in that region, and consequently the increase in the percentage of profits from this product.
- o Saving time and effort for the consumer by knowing the exact time of arrival of the product to the place of consumption.

1.8 Research Structure

This research consists of the following chapters:

Chapter1: Contains the overview, definition, problem, objectives, importance of the project, scope, expected outcomes and research structure.

Chapter2: Talks about literature survey related to the research issue.

Chapter3: Talks about research methodology.

Chapter4: Talks about system analysis, tools and techniques used in.

Chapter5: Explain the implementation (how the system works).

Chapter6: Talks about recommendations and the conclusion.

CHAPTER TWO

LITERATURE SURVEY

1. Introduction

This chapter talks about the field of the logistics service and previous studies associated with this research.

2.1.1. Smart Logistic Services System

It's way obvious that the process of transporting products and food is done traditionally consuming a lot of time and efforts along with the high possibility of damaging or ruining the transported product leading to budget consumption. Based on comes the need of developinng a smart logistic services system so that the process of transporting goods or products from the place of production to the consumer is done in a smart and smooth way by using digital maps. Thus, the required time, effort and cost are saved for both the producer and the consumer.

2.2 Previous Studies

2.2.1 Researching the effect of logistics on the competitive advantages of container transport in Port Sudan port (#1)

The research is an applied study to measuring rates, and studying research hypotheses for the purpose of analysis and comparison of some models applied in neighboring ports. The importance of the research lies in the scientific and practical importance of logistics and its applications, due to its implications for the national economy, and the research came out with several results, the most important of which are: Bahri towards the hub ports because of its great benefits, the most important of which is speed and reducing the cost of freight, and the emergence of alliances and conglomerations between shipping lines and international port management companies [2].

2.2.2 Ph.D Research on the effect of applying the logistical management approach on the performance of industrial production institutions in Sudan, 2000-2009 (#2)

In the absence of logistical systems in most Sudanese industries this research was applied aiming to identify the topic of logistics management and its applications to the performance of

industrial production enterprises in Sudan. The general objective of the study is to review the current situation of the Sudanese industry and the bottlenecks it faces in the application of the logistical management approach. The special aim is to identify the effect of applying the logistical management approach in the Giad Holding Company. The importance of the study lies in the scarcity and lack of researches and studies in this field [3].

2.2.3 The use of information technology in logistical activities and its impact on achieving a competitive advantage for the organization (#3)

The aim of this study was to identify the impact of information technology related to logistics on modifying the effect relationship between the logistics and the competitive advantage of the pharmaceutical complex of SAIDAL.

The dimensions of the independent variable were the effectiveness of logistics activities, the internal integration of logistics activities and the adoption of the philosophy of supply chains. the study concluded with a number of results, and the important one is: - The pharmaceutical complex is concerned with the management of logistics activities effectively [4].

2.2.4 Halal Economics (#4)

Halal industry is one of the fastest growing sectors in the world. the Halal industry has now expanded beyond the food sector to include pharmaceuticals, cosmetics, health products, medical supplies and devices, as well as service sector components such as logistics, marketing, print and electronic media, Packaging, branding, and financing. the use of halal products and their spread and effects on the material and spiritual life of Muslims and the dimensions of the term halal jurisprudence and religion on the one hand and the need to benefit from modern studies and modern technology in the production and monitoring of Halal and its products on the other hand, as well as the development of exports and imports of various Halal products in the field International trade and economy [5].

2.2.5 Integrating BIM and GIS to improve the visual monitoring of construction supply chain management CSCM (#5)

This paper integrates building information modeling (BIM)

and geographic information systems (GIS) into a unique system, which enables keeping track of the supply chain status and provides warning signals to ensure the delivery of materials. First, the proposed methodology is implemented by using BIM due to its capability to accurately provide a detailed takeoff in an early phase of the procurement process. Furthermore, in order to support the wide range of spatial analysis used in the logistics perspective (warehousing and transportation) of the CSCM, GIS is used in the present model.

Thus, this paper represents the integrated GIS-BIM model manifesting the flow of materials, availability of resources, and “map” of the respective supply chains visually [6].

A table below that shows the similarities and differences between previous studies and current study:

Table 2.1 comparison between previous and current study

Study	Similarities	Differences
#1	Using logistics and its applications	In our smart logistic services system, we attempt to use GIS and smart network maps
#2	Using logistics and its applications	In our smart logistic services system, we attempt to use GIS and smart network maps
#3	Using logistics and information technology and supply chains	In #3 the logistics and supply chains were used to alongside to help in the management of the organization. In our smart logistic services system, we attempt to use all of that besides GIS and smart network maps to help the distribution of products
#4	Using logistics and information technology and supply chains	The economics of the halal market #4 use the concept of logistics services by studying the groups of society to identify Muslim groups in society to provide services compatible with the religion and beliefs of Muslims. In our smart logistic services, it aims to study the community and provide appropriate and required services and provide the necessary needs for each region.

#5	Using logistics and information technology and GIS	In #5 integrates building information modeling (BIM) and geographic information systems (GIS) into a unique system, which enables keeping track of the supply chain status and provides warning signals to ensure the delivery of materials. In our smart logistic services it aims to study the community and provide appropriate and required services and provide the necessary needs for each region using the GIS and smart network maps.
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2.3 TOOLS AND TECHNIQUES

2.3.1 UML

Unified Modeling Language, or UML, is a visual language that helps software developers visualize and construct new systems. It's not a programming language it's a set of rules specifically for drawing diagrams. There are many types of software engineering diagrams, but think of this language like the software engineer's version of a blueprint.

UML diagrams are one of the most popular software engineering diagrams. When it comes to building new products or systems, there are two key reasons to add UML diagramming to your toolkit.

Before developers start coding, UML diagrams can help everyone get on the same page. Understanding the system, they're trying to create allows developers to delegate work, identify potential problems before work has started, and work efficiently toward a common goal.

After code is written, a UML diagram can help developers understand the decisions made and structures developed for the project. This information helps teams as they look to improve upon the project for the future [7].

Use Case Diagram

A use case diagram is a way to summarize details of a system and the users within that system. It is generally shown as a graphic depiction of interactions among different elements in a system. Use case diagrams will specify the events in a system and how those events flow, however, use case diagram does not describe how those events are implemented [8].

Sequence Diagram

Sequence diagrams are a popular dynamic modeling solution in UML because they specifically focus on lifelines, or the processes and objects that live simultaneously, and the messages exchanged between them to perform a function before the lifeline ends. Along with our UML diagramming tool, use this guide to learn everything there is to know about sequence diagrams in UML [9].

Activity Diagram

An activity diagram is essentially a flowchart that shows activities performed by a system.

Activity diagrams, along with use case and state machine diagrams, are considered behavior diagrams because they describe what must happen in the system being modeled.

Stakeholders have many issues to manage, so it's important to communicate with clarity and brevity. Activity diagrams help people on the business and development sides of an organization come together to understand the same process and behavior [10].

Class Diagram

One of the more popular types in UML is the class diagram. Popular among software engineers to document software architecture, class diagrams are a type of structure diagram because they describe what must be present in the system being modeled. No matter your level of familiarity with UML or class diagrams, our UML software is designed to be simple and easy to use.

UML was set up as a standardized model to describe an object-oriented programming approach. Since classes are the building block of objects, class diagrams are the building blocks of UML. The various components in a class diagram can represent the classes that will actually be programmed, the main objects, or the interactions between classes and objects [11].

2.3.2 Entity Relationship Diagram(ERD)

An Entity Relationship (ER) Diagram is a type of flowchart that illustrates how “entities” such as people, objects or concepts relate to each other within a system. ER Diagrams are most often used to design or debug relational databases in the fields of software engineering, business information systems, education and research. Also known as ERDs or ER Models, they use a defined set of symbols such as rectangles, diamonds, ovals and connecting lines to depict the interconnectedness of entities, relationships and their attributes. They mirror grammatical structure, with entities as nouns and relationships

as verbs [12].

2.3.3Hyper Text Markup Language (Html)

HTML stands for Hyper Text Markup Language. HTML is a markup language for describing web documents (web pages) [13].

2.3.4 Cascading Style Sheet (CSS)

CSS stands for Cascading Style Sheets. CSS defines how HTML elements are to be displayed [14].

2.3.5Bootstrap4

Bootstrap is a free front-end framework for a faster and easier web Development. Bootstrap includes HTML and CSS-based design templates for Print, forms, buttons, tables, navigation, media, image swirls, and many other components, plus optional JavaScript plug-ins.

Bootstrap also gives you the ability to easily create responsive designs [15].

2.3.6 Laravel8

Laravel is a free and open source PHP web framework created by Taylor Otwell is intended to develop the following web applications Architectural Model Display Controller (MVC) is based on Symfony [16].

2.3.7VISUAL STUDIO CODE

Microsoft Visual Studio is an integrated development environment (IDE) from Microsoft. It is used to develop computer programs, as well as websites, web apps, web services and mobile apps. Visual Studio uses Microsoft software development platforms such as Windows API, Windows Forms, Windows Presentation Foundation, Windows Store and Microsoft Silverlight. It can produce both native code and managed code [17].

2.3.8 Flutter Framework

Flutter is a free and open-source mobile UI framework created by Google. In a few words, it allows you to create a native mobile

application with only one codebase. This means that you can use one programming language and one codebase to create two different apps (for iOS and Android) [18].

2.3.9 Android Studio

Android Studio is the official integrated development environment (IDE) for Android application development. It is based on the IntelliJ IDEA, a Java integrated development environment for software, and incorporates its code editing and developer tools.

To support application development within the Android operating system, Android Studio uses a Gradle-based build system, emulator, code templates, and GitHub integration. Every project in Android Studio has one or more modalities with source code and resource files. These modalities include Android app modules, Library modules, and Google App Engine modules [19]

2.3.10 ArcGIS

Is a **GIS** (geographical information systems) software which allows users to manage and analyze geographical information by visualizing a range of geographical statistics through layer building maps (such as trade flows or climate data). It is used by a whole host of academic institutions and departments, both in the sciences and humanities, to develop and illustrate groundbreaking research. Further it is used by many governments and commercial/private institutions worldwide [20].

2.4 Summary

Developing a smart logistic services system will help finding a solution to the problem of product distribution thus helping companies to produce the largest possible amount of products and distribute them in the best way, increasing the profits of the companies and creating a competitive advantage for the companies providing the distribution service, and gaining the confidence and satisfaction of customers. In the next chapter the methodology of the research will be explained.

CHAPTER THREE

RESEARCH METHODOLOGY

2

3 Introduction

This chapter explains various methodologies that were used in gathering data and analysis which are relevant to the research. The methodologies will include areas such as the Area of Study, research design, Research Approach, data collection method and its management.

4 Research Design

The research design for this study was defined as a descriptive research method, describing the characteristics of logistics services. Data and case study requirements for this research (Dal Food Company) were collected through interviews, questionnaires and observation, and previous studies were collected and analyzed to find out similarities and differences.

The data collected from the case study was analyzed to find out the problems facing the distribution operations and find feasible solutions to them.

This design always places more emphasis on research subject and not the reason for the research subject.

5 Research Approach

The research method is descriptive, analytical and applied scientific qualitative approach, hence the assumption was using information technology and smart maps system will improve the activity of shipping of the products which requires contextual data, and because of its advantages and analysis of data and determining the nature of the problem and setting questions and testing the validity of the assumptions.

6 Data Collection

In this study, the researcher will use questionnaires, interviews and observation as a way of data collection.

6.1.1 Interview

We use this method of collecting data through oral and verbal communication between the researcher and the respondent, Both structured and unstructured questions, interview was used because

they are quite flexible, adaptable and can easily allow us to see to get more insight on the topic. It was used to find views of Senior Manager and Manager of Dal Foot on the performance of the prepaid metering system.

6.1.2 Observation

In our research, we use an observer method as a participant to collect data by watching behavior or events or a note from the initial stage, which is the shipment of products or commodities to the distribution vehicles and their exit from the place of production or factory and the way they move on the roads until they reach the distribution line and to the points of distribution and arrival to the last point and then return to the place of production or the factory again.

7 Data Analysis

The data collected through the smart logistic services system is analyzed to predict the factors that help the producing companies to make administrative decisions that will increase their profits and financial returns, and also to know the most requested products in each region, in order to distribute the right product in the right place.

8 Case study

8.1.1 Introduction

The study was carried out in Khartoum city, taking Dal groups for food industries as case study. The rationale behind this choice is from the sense that Dal groups is the biggest and most dominated company in food industries sector besides the large experience they have in the field. Hence, there is a possibility of obtaining all necessary and relevant data from the company.

8.1.2 Scenario

The process of the distribution depends on an important factor which is the ordering of the products, they are two types of orders:

- **Pre-order**

In this type of ordering the clients orders a specified quantity of the product to be delivered.

Two types of clients can be declared:

a) Near course:

In this case the distributor distributes the product directly to the client.

b) Far course:

The distributor distributes the product to the client's closest relied agent to deliver it.

- **Random sale**

In this type the distributor packs an appropriate quantity of products and distribute it randomly to the distribution points.

8.1.3 Challenges

1. High cost (there is no optimal paths)
2. Time and effort lost (absence of monitoring)
3. Misuse of resources (no fully final report)
4. There is no use of the collected data (wasted data)

8.1.4 Solutions

1. Provide the optimal paths through the digital maps.
2. Provide monitoring over the distribution process using real time location service.
3. Provide full final report (the payment, the quantity sold, the reminder, time of ordering and time of delivery).
4. Use of the collected data in future marketing decisions through the Business Intelligence (BI).

9 Summary

This chapter clarifies the important points represented in the method used in the research design and the research approach.

In addition to clarifying how to collect data and analyze this data, a

description of the case study is the methodology that this research follows. The next chapter will discuss system analysis and design.

CHAPTER FOUR

SYSTEM ANALYSIS AND DESIGN

4.1 Introduction

This chapter explains and illustrates of the system description, design, function and system analysis using the unified modelling language (UML) diagrams and the database analysis.

4.2 System description

This project is a system that manages the process of distributing products from the place of storage to the place of consumption by providing smart digital maps as well as providing control over the distribution process using real-time tracking techniques and visualize data.

4.3 System Architecture

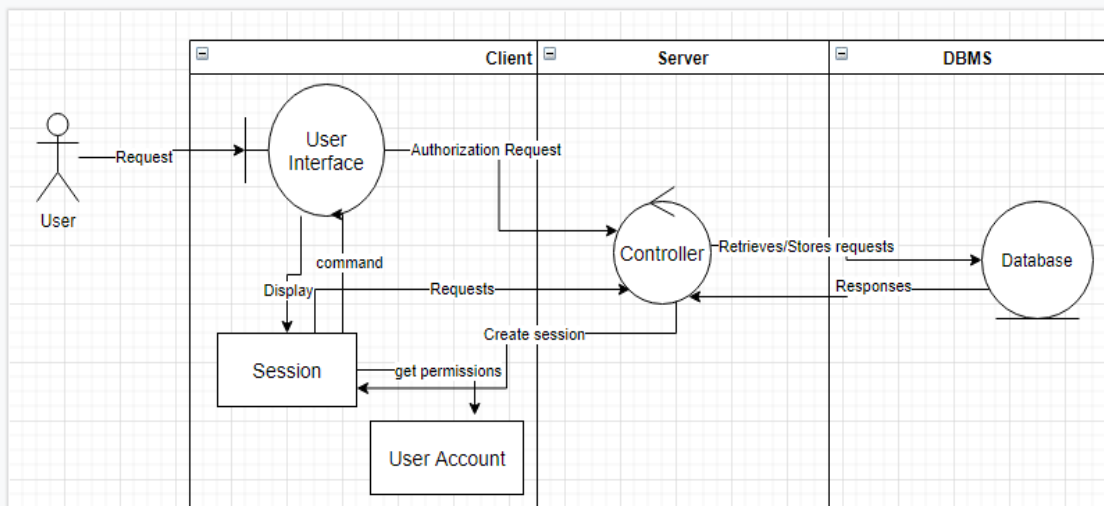


Figure 4.1 System Architecture Diagram

User: any person who interacts with the system whether admin or distributor.

User Interface: The interface in which used by the user to communicate with the system.

Clients: a computer hardware or software that request resources and services made available by server.

Controller: An intermediary that validate users' information and allows/prevents data access, retrieves and stores data to and from the database.

Server: a computer program or device that provides a service to another computer program and its user.

Session: The point of contact for all browser requests for users that are logged-in and authenticated.

User Account: Contains data retrieved from the DBMS specifying the user's access level and permissions.

DBMS: software designed to store, retrieve, define, and manage data in a database.

Database: an organized collection of structured information, or data, typically stored electronically in a computer system.

4.4 System Requirement:

This section explains the system requirements.

4.4.1 Functional Requirement

1. **Create Account:** The admin and the distributor must create an account by registering the required information, and then log in to the system to access the information or perform their tasks.
2. **Login to System:** The admin and distributor must log in to the system first by entering the required information and then carry out their tasks.
3. **Activate distributor account:** The admin must be able to activate the distributor's account to be able to log in to the system.
4. **Update distributor account:** The admin must update the distributor's account information when there is a change in it.
5. **Deactivate distributor account:** The admin must be able to

deactivate the distributor's account if he no longer employed for any reason illness, suspension, or death.

- 6. View all distributors accounts:** The admin can view the information of all distributor accounts registered in the system.
- 7. Monitor and Track the distribution:** The admin must be able monitor and track the distribution process in real time through smart digital maps.
- 8. Add products:** The admin must be able add products to the system if there are new products produced.
- 9. Update product's prize and amount:** The admin must be able to update the price of the product or its amount if there is a change in the product's price or its available amount.
- 10.Data Visualization:** The system should provide reports about the distribution's processes by visualizing the data as heat maps, cluster maps and charts.
- 11.Add distribution points:** The distributor must be able to add the distribution points that must be covered and add the new distribution points that showed up in his distribution path.
- 12.Manage distribution points:** The distributor must be able to manage the distribution points by evaluating them, update, or deleting them if they are no longer exist.
- 13.Order products:** The distributor must be able order the amount of products he must distribute at his distribution points
- 14.View all sales:** The distributor can display all the products that he sold in the distribution points he covers on the map.

4.4.2Non-functional Requirement

4.4.2.1 Performance Requirements

- The system should have accessed and be usable within 10 seconds.
- The system should update the interface on interaction within 10 seconds as maximum.
- The database should be normalized to prevent redundant data and improve performance.

4.4.2.2 Safety and Security Requirements

4.4.2.2.1 Safety Requirements

Backups of the databases should be done weekly and be kept for one month

4.4.2.2.2 Security Requirements

Only the intended people can access to the database and update or delete it information

4.4.2.2.3 Software Quality Attributes

Correctness: the application should never allow anyone to read information not intended for that person

Maintainability: the application should use continuous integration so that features and bugs fixes can be deployed quickly without downtime

Usability: the interface should be easy to use without a tutorial and allow users to accomplish their goals without errors

4.5 System Analysis & Design

This section describe how system was analyzed using UML (Unified Modeling Language).

4.5.1 Use Case Diagram

The figure below shows the use case diagram of the system which consists of the admin and the distributor as actors performing the system functions.

The functions performed by the admin:

1. Registration
2. Login
3. Monitoring and tracking the process of distribution.
4. Managing the distributors registered on the system and this process is consisting of:
 - o Activate distributor account.
 - o Deactivate distributor account.
 - o View all distributors registered on the system.
 - o Deals with the reports generated by the system.
5. Managing products and this process is consisting of:
 - o Update products price
 - o Update products amounts
 - o View all the products

The functions performed by the distributor:

1. Registration
2. Login
3. Managing distributions point and this process is consisting of:
 - o Add distribution points
 - o Deleting distribution points
 - o Update distribution points information
 - o View all the distribution points and location of each one.

6. Ordering process and its consisting of:

- o Order products
- o Sale products
- o View sales

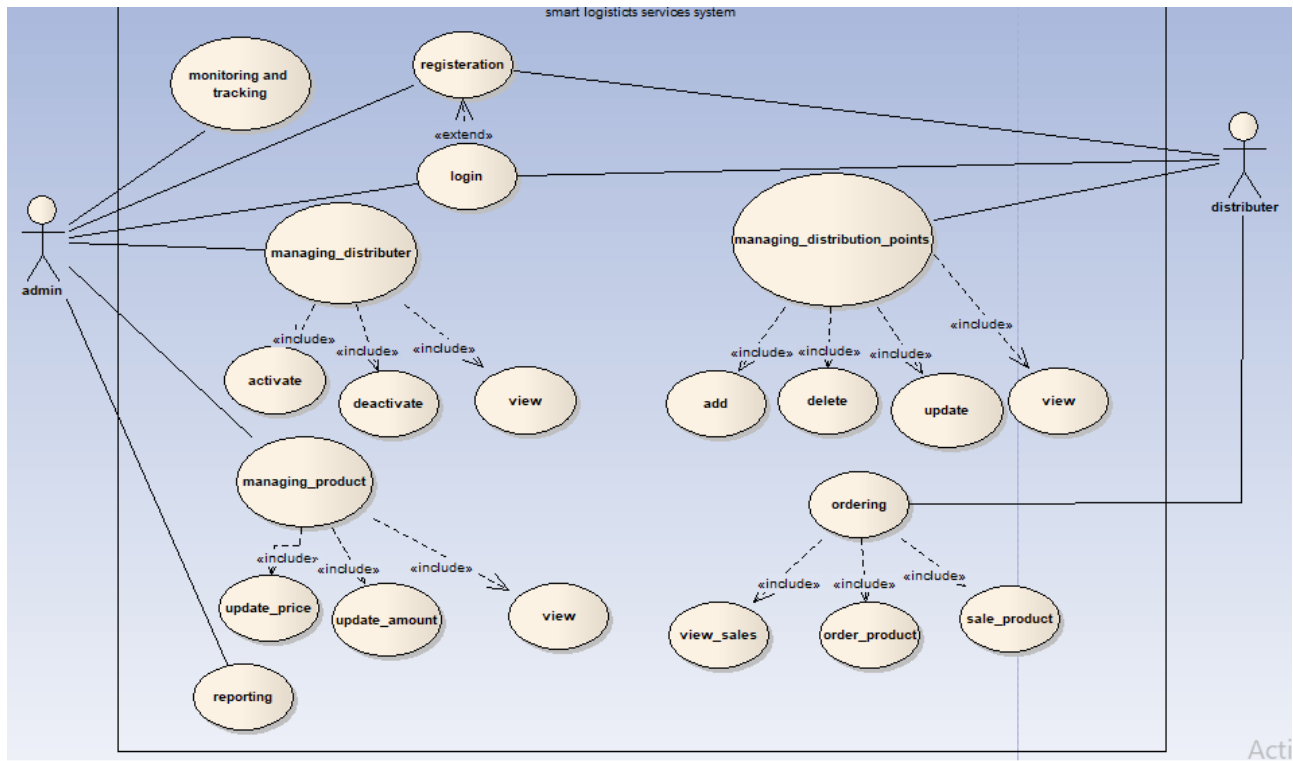


Figure 4.2 Use Case Diagram

4.5.2 Sequence Diagrams

4.5.1.2 Admin sequences diagrams

- **Registration:**

The figure below illustrates the process of admin registration which done when admin enters the required information to register to the system and be able to log in to the system.

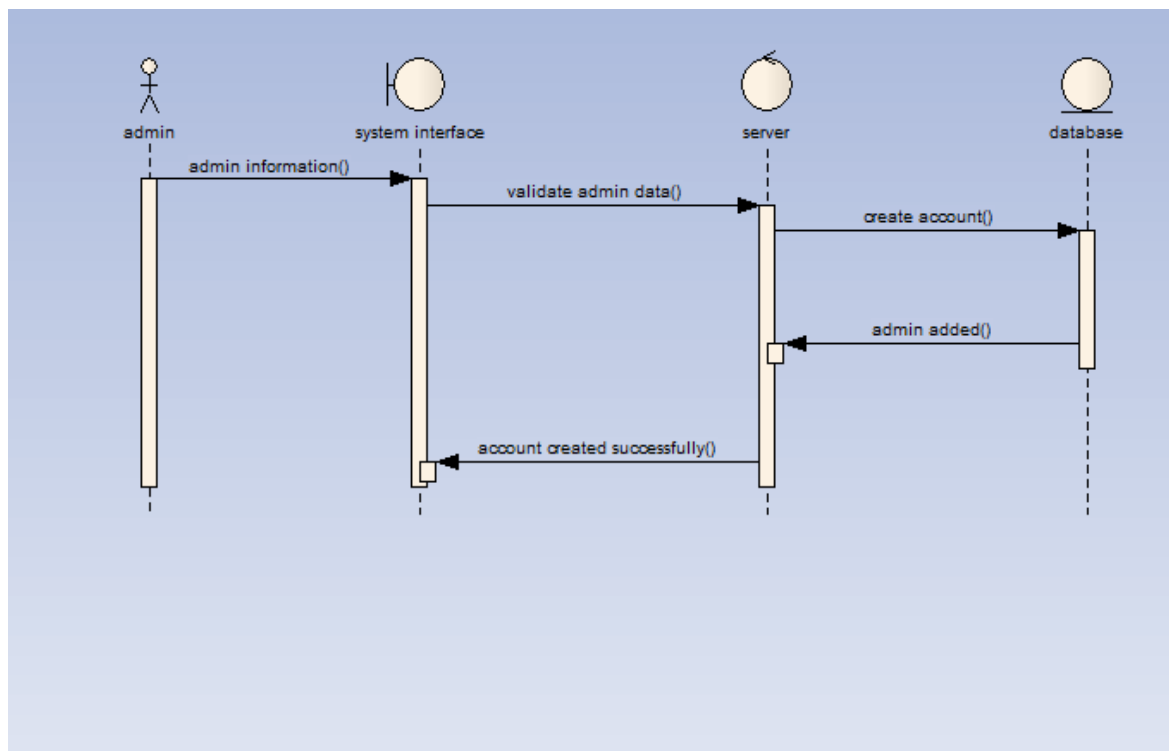


Figure 4.3 Sequence Diagram for admin registration

- **Login:**

The figure below illustrates the process of admin login which done when admin enters the required information and validate this information to ensure whither he has account or not to give him permission to access the system or denied his access to the system.

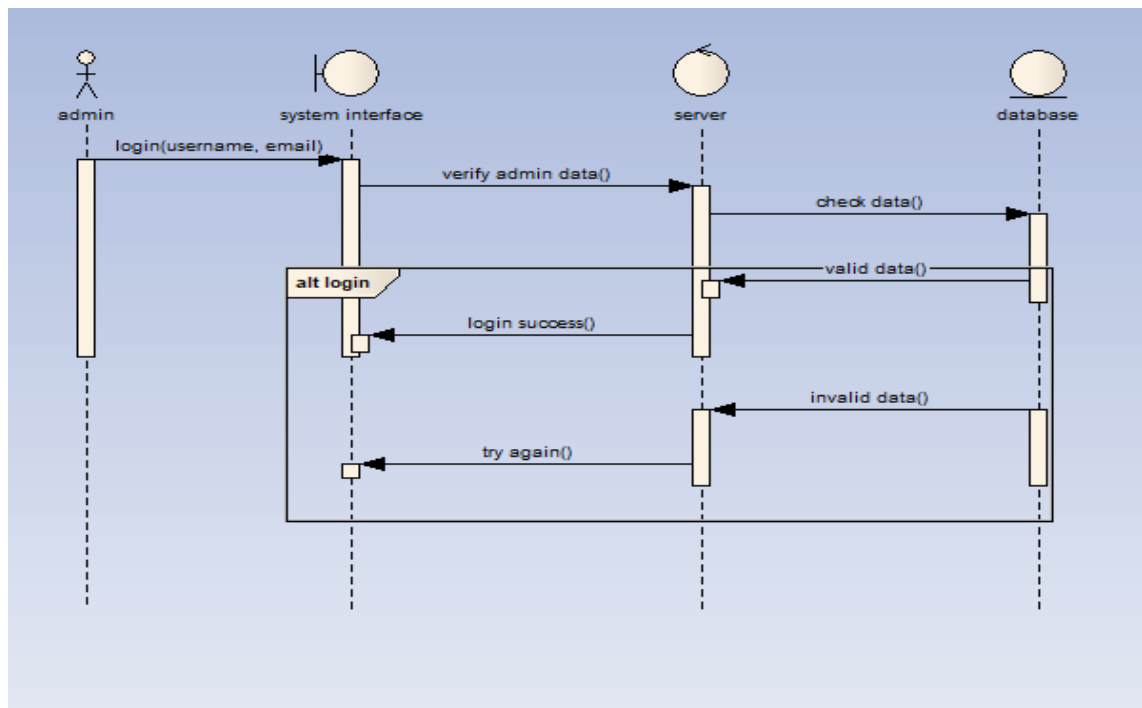


Figure 4.4 Sequence Diagram for admin login

- **Activate:**

The figure below illustrates the process of activating distributor account which require the registration of the distributor first then the admin activate the distributor account to enable him to log in to the application.

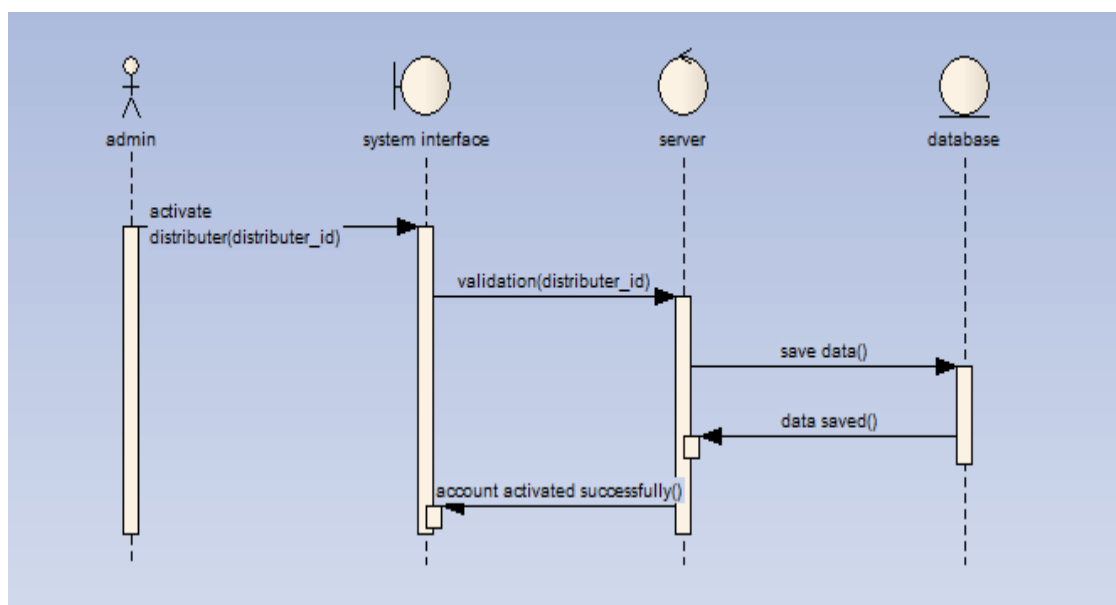


Figure 4.5 Sequence Diagram for activating distributor account

- **Deactivate:**

The figure below illustrates the process of deactivating distributor account and its done by the admin when the distributor violates the producing company policies or due to illness or death of the distributor.

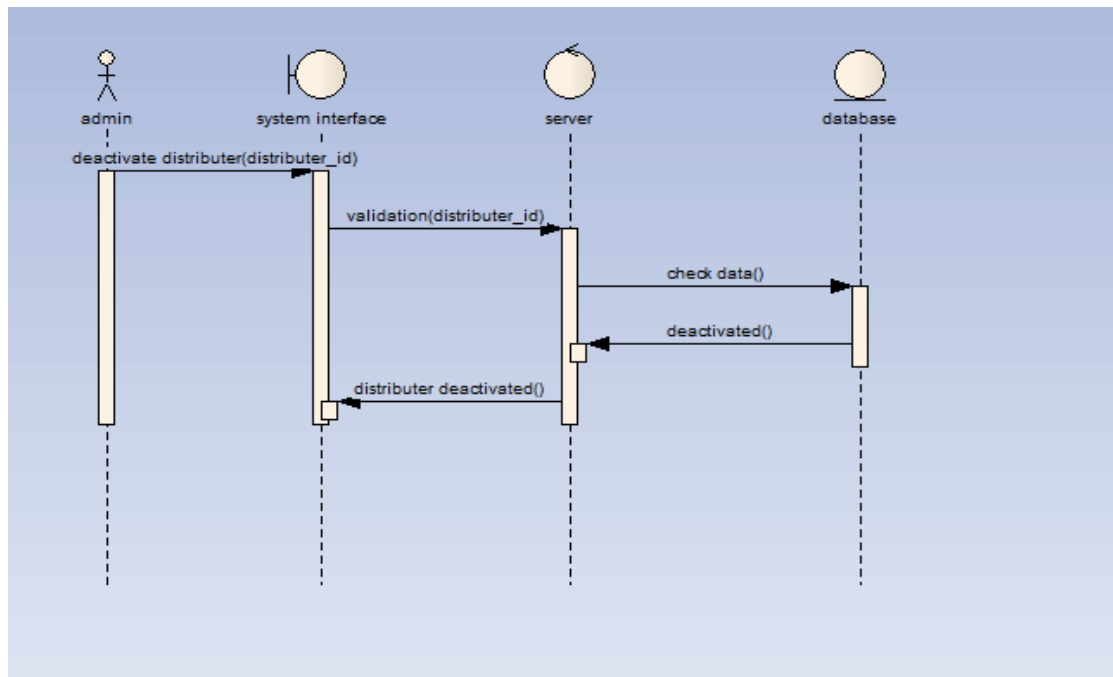


Figure 4.6 Sequence Diagram for deactivating distributor account

- **View**

The figure below illustrates the process of displaying all distributors accounts registered to the system.

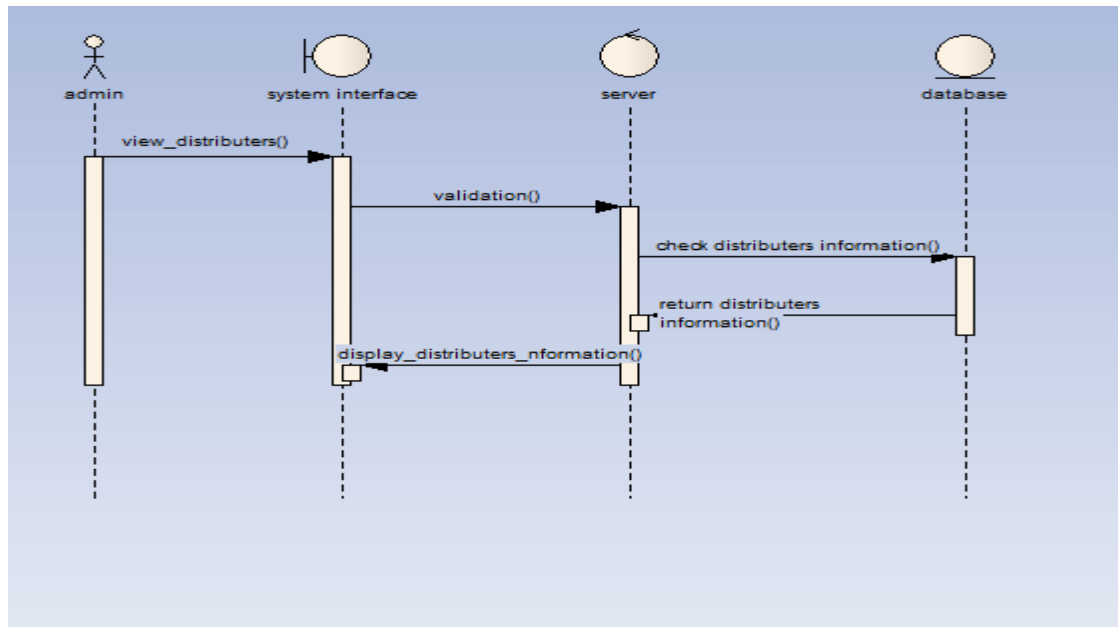


Figure 4.7 Sequence Diagram for viewing distributers information

- Monitoring and Tracking**

The figure below illustrates the process of monitoring and real time tracking of a distribution process performed by a specific distributor, this process is done by the admin checking the current distribution point the distributor covers at that period of time when the process of monitoring took place and the information of that distribution point, also determine the distributor location.

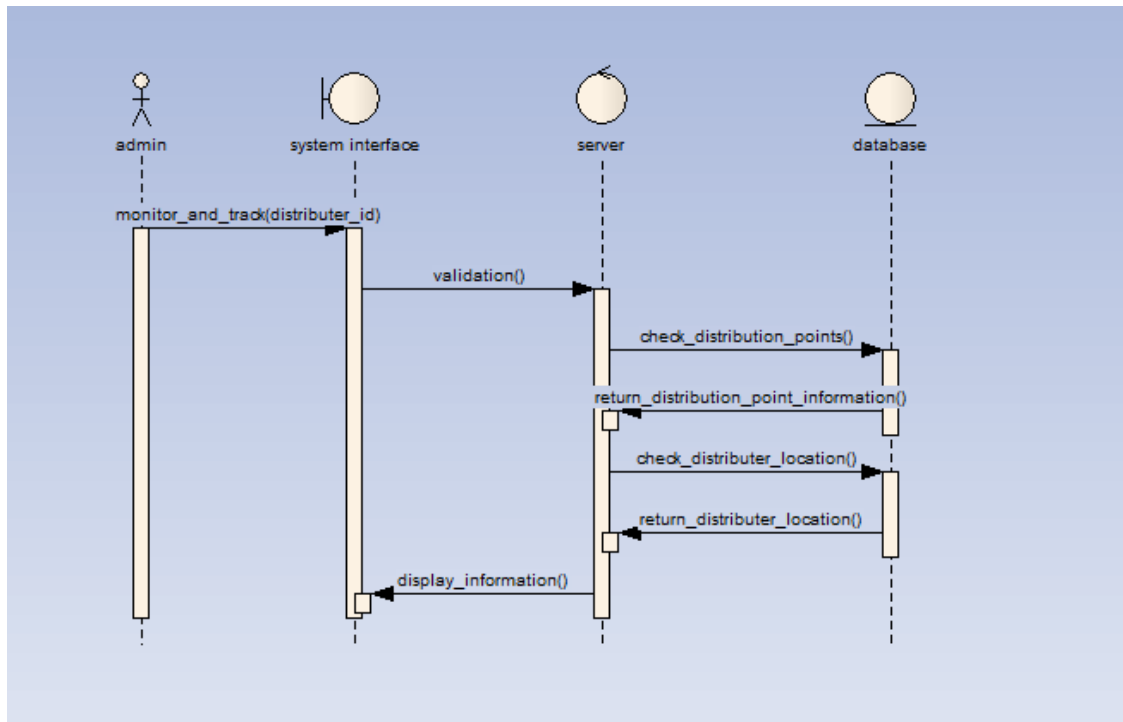


Figure 4 8 Sequence Diagram for monitoring and tracking of distribution process

- **Reporting**

The figure below illustrates the process of the admin dealing reports generated by the system in order to use these reports in order to improve the decisions making process.

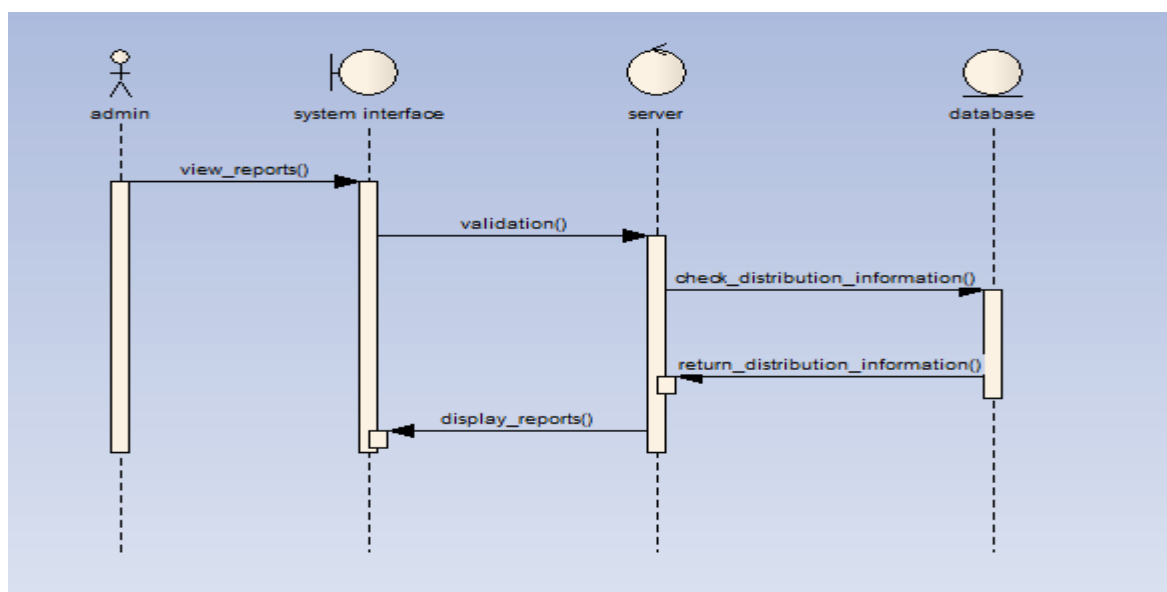


Figure 4.9 Sequence Diagram for reporting

- **Updating products price**

The figure below illustrates updating the prices of the products done by the admin.

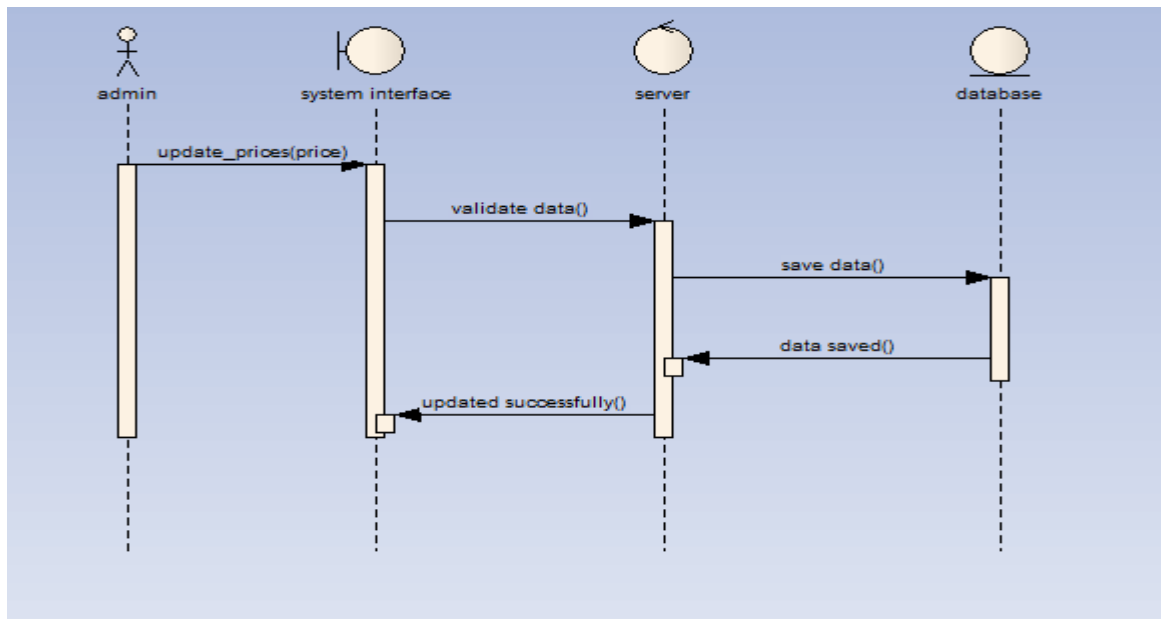


Figure 4.10 Sequence Diagram for updating product price

- **Updating products amount**

The figure below illustrates updating the amounts of the products done by the admin.

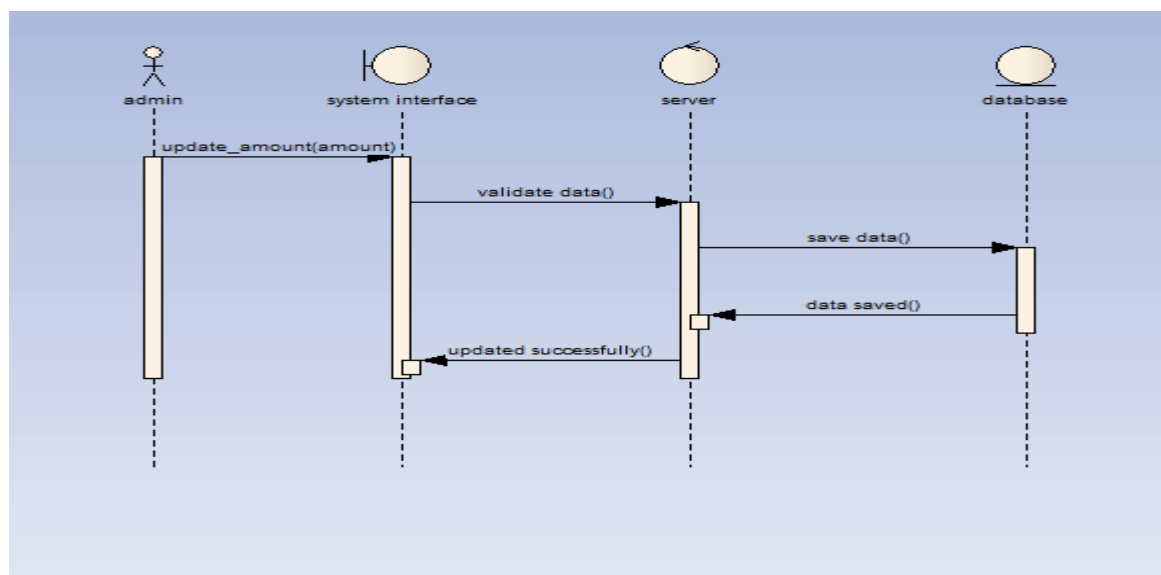


Figure 4.11 Sequence Diagram for updating product amount

- **View products**

The figure below illustrates the process of viewing all the products in the database of the system, their names, description, prices, and all other information related to the product.

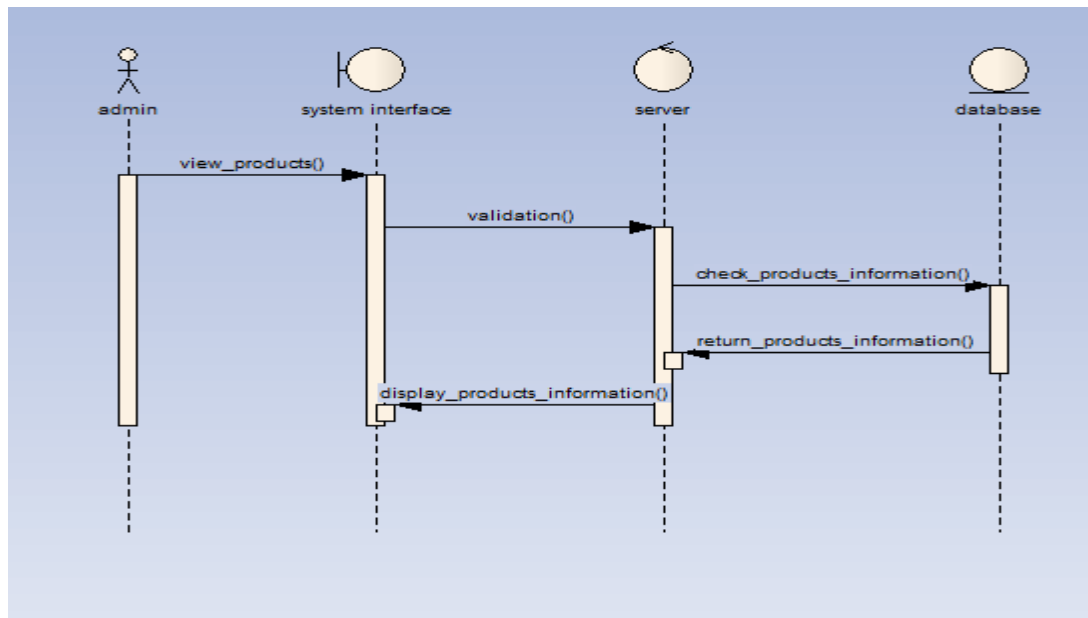


Figure 4.12 Sequence Diagram for view products

4.5.1.3 Distributer sequences diagrams

- Registration

The figure below illustrates the process of distributer registration which done when distributer enters the required information to register to the system and be able to log in to the application.

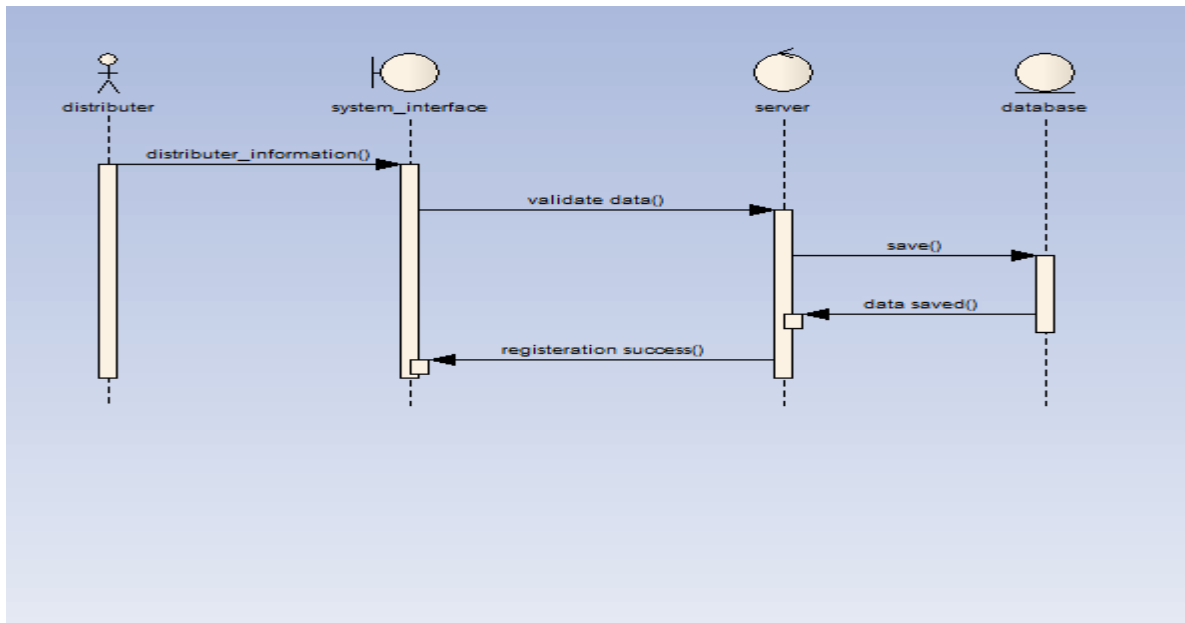


Figure 4.13 Sequence Diagram for distributer registration

- **Login**

The figure below illustrates the process of distributor login which done when distributor enters the required information and validate this information to ensure whither he has account or not to give him permission to access the application or denied his access to the system.

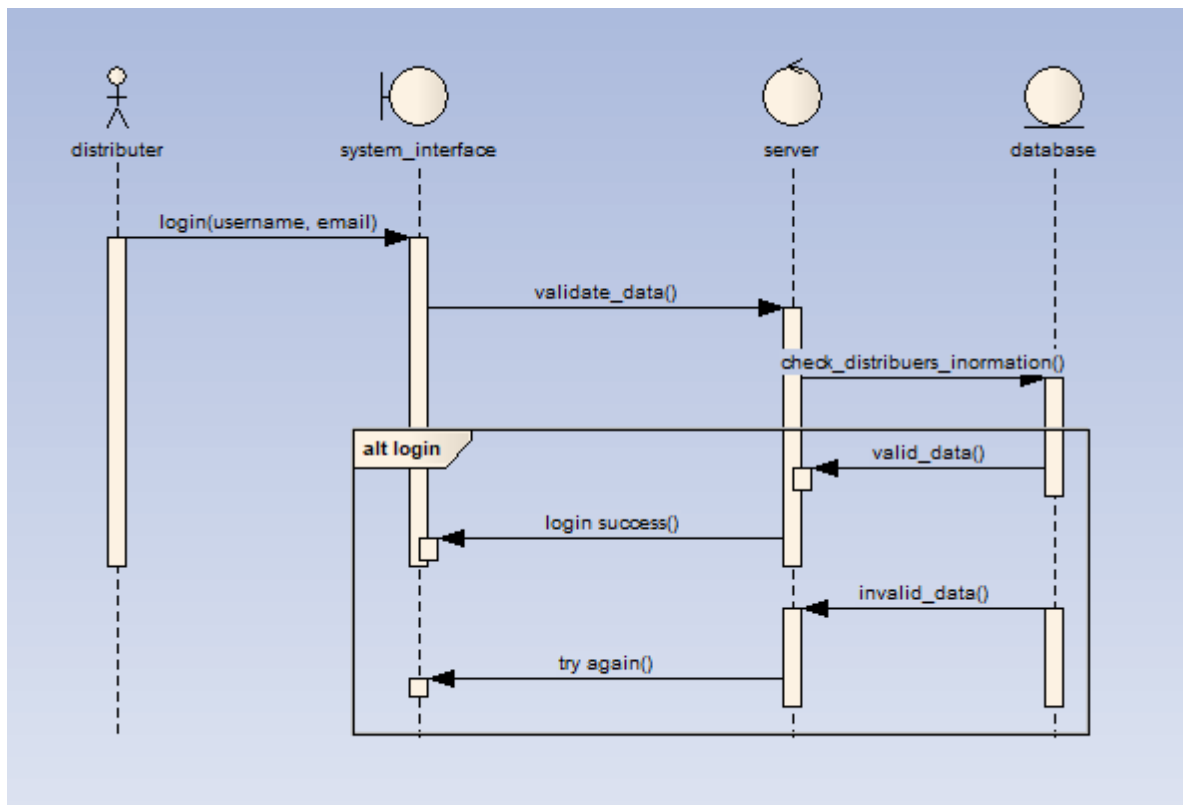


Figure 4.14 Sequence Diagram for distributor login

- **Adding new distribution points**

The figure below illustrates the process of adding new distribution point which done by the distributor when he discovers a new one on the distribution area he covers, the distributor registers the point by entering the information of that point and its location.

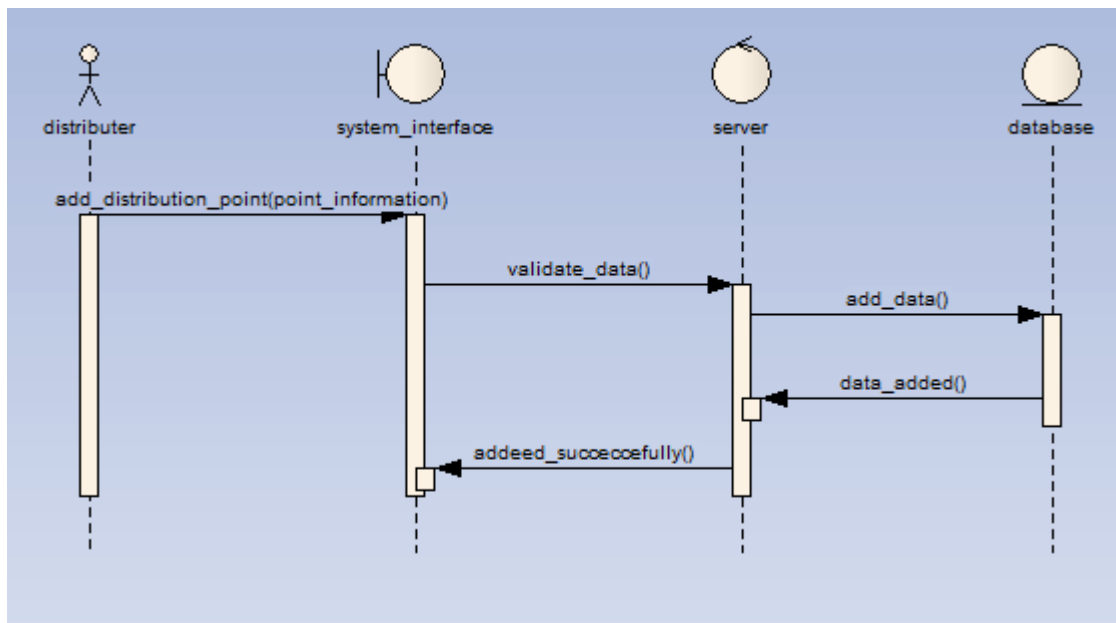


Figure 4.15 Sequence Diagram for adding new distribution point

- **Deleting distribution points**

The figure below illustrates the process of deleting a distribution point which done by the distributor when this point has been no longer available due to any reasons.

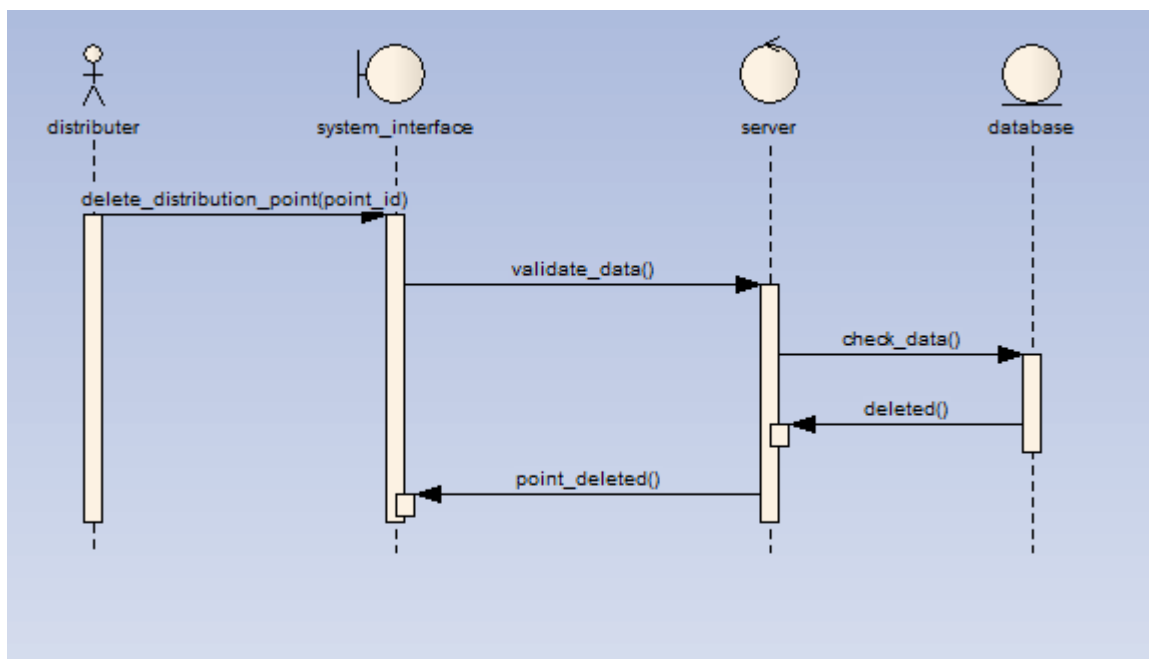


Figure 4.16 Sequence Diagram for deleting distribution point

- **Updating distribution points information**

The figure below illustrates the process of updating the information of a distribution point.

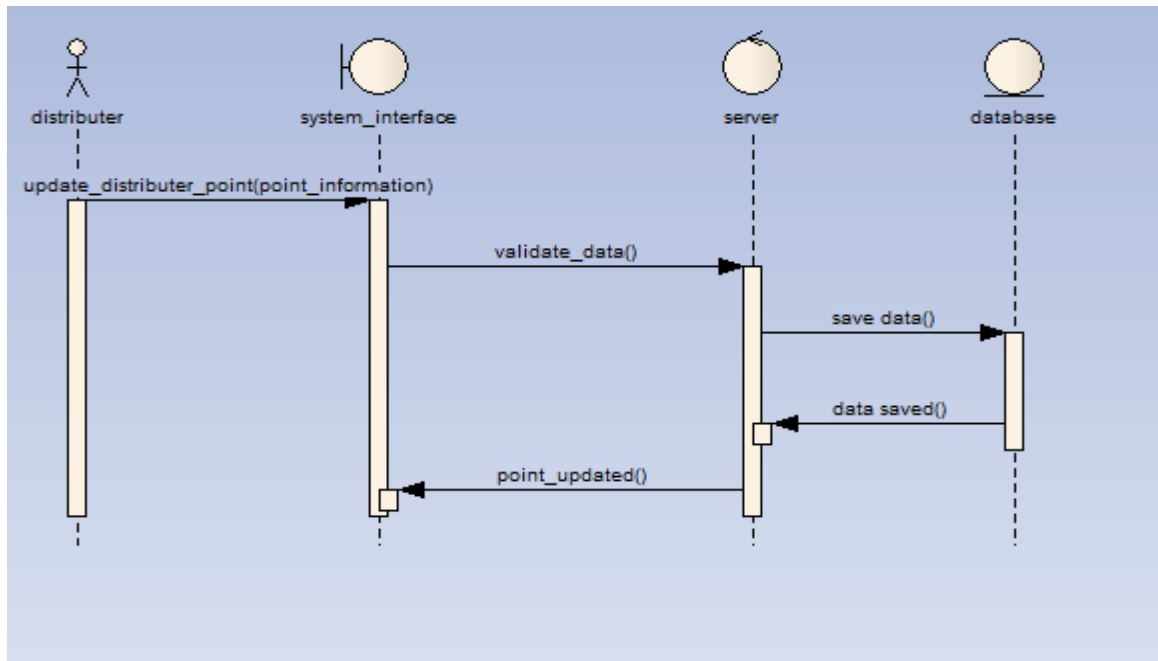


Figure 4. 17 Sequence Diagram for updating distribution point

- **Viewing distribution points**

The figure below illustrates the process done by the distributor viewing all the distribution points of a distribution area that he covers.

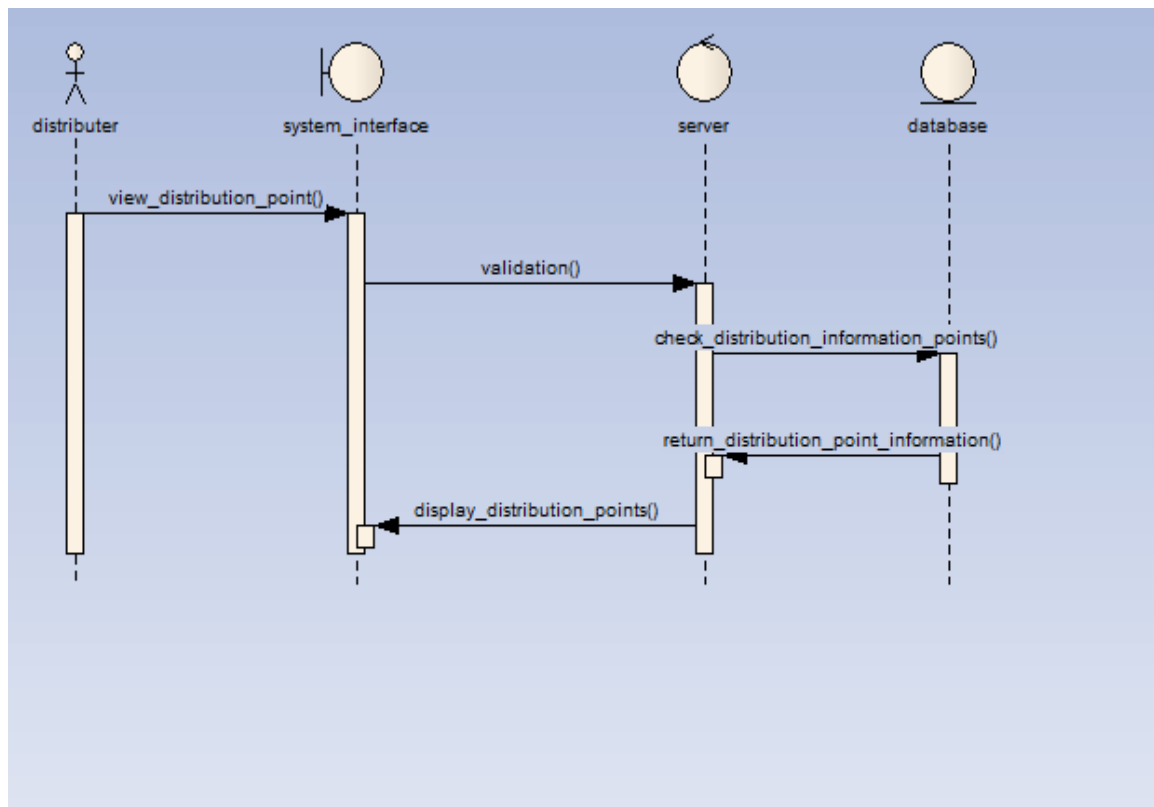


Figure 4.18 Sequence Diagram for viewing distribution point

- **Ordering products**

The figure below illustrates the distributors ordering process for products, if the amount requested is available the order is accepted if not the order is rejected and the distributors is informed that there is no available amount.

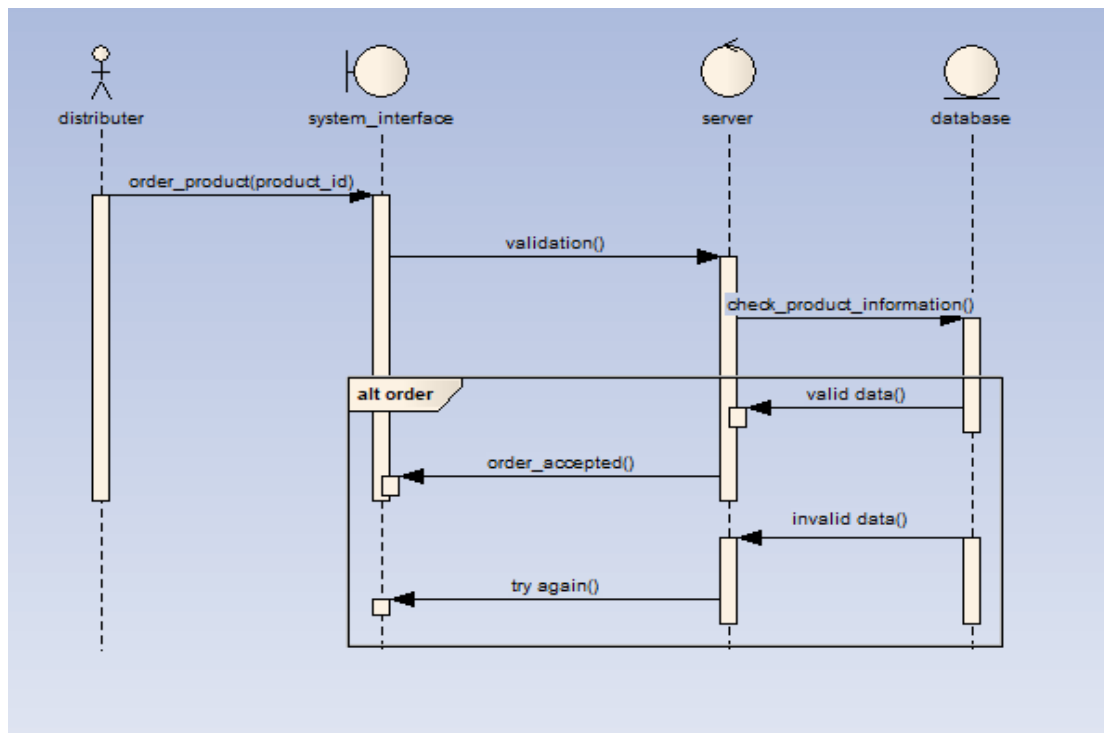


Figure 4.19 Sequence Diagram for ordering product

- **Sale products**

The figure below illustrates the process of selling products by the distributors, if the distributor is in determined range of the distribution point (200-meter radius) the sale is performed otherwise the operation fails.

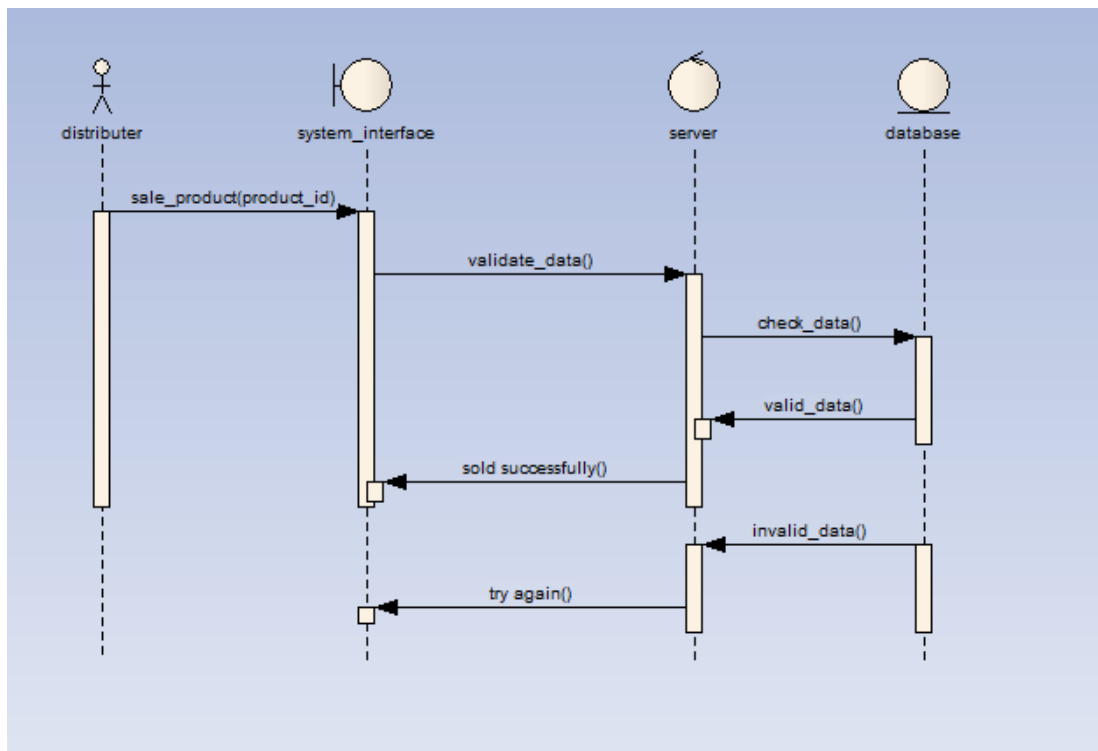


Figure 4.20 Sequence Diagram for sale products

- **View sales**

The figure below illustrates the process the distributors viewing the sales operations they have performed.

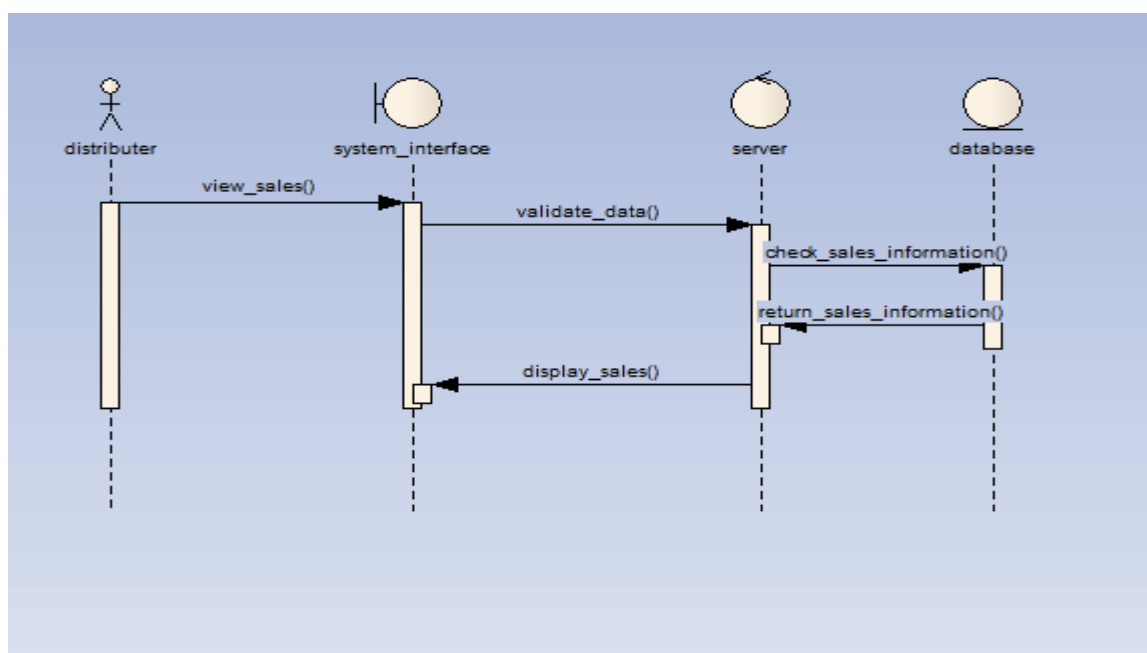


Figure 4.21 Sequence Diagram for view sales information

4.5.3 Activity Diagram

The figure below illustrates the activity diagram of the system for both the admin and the distributor.

The initial process for both the admin and the distributor is the registration process if they have no account in the system, if they have accounts the initial process would be the login, if the login was successful the other processes would be proceeded for otherwise the system well forward back to login.

After the successful login the admin and the distributor would be able to perform their tasks and functions illustrated before in the sequence diagrams section.

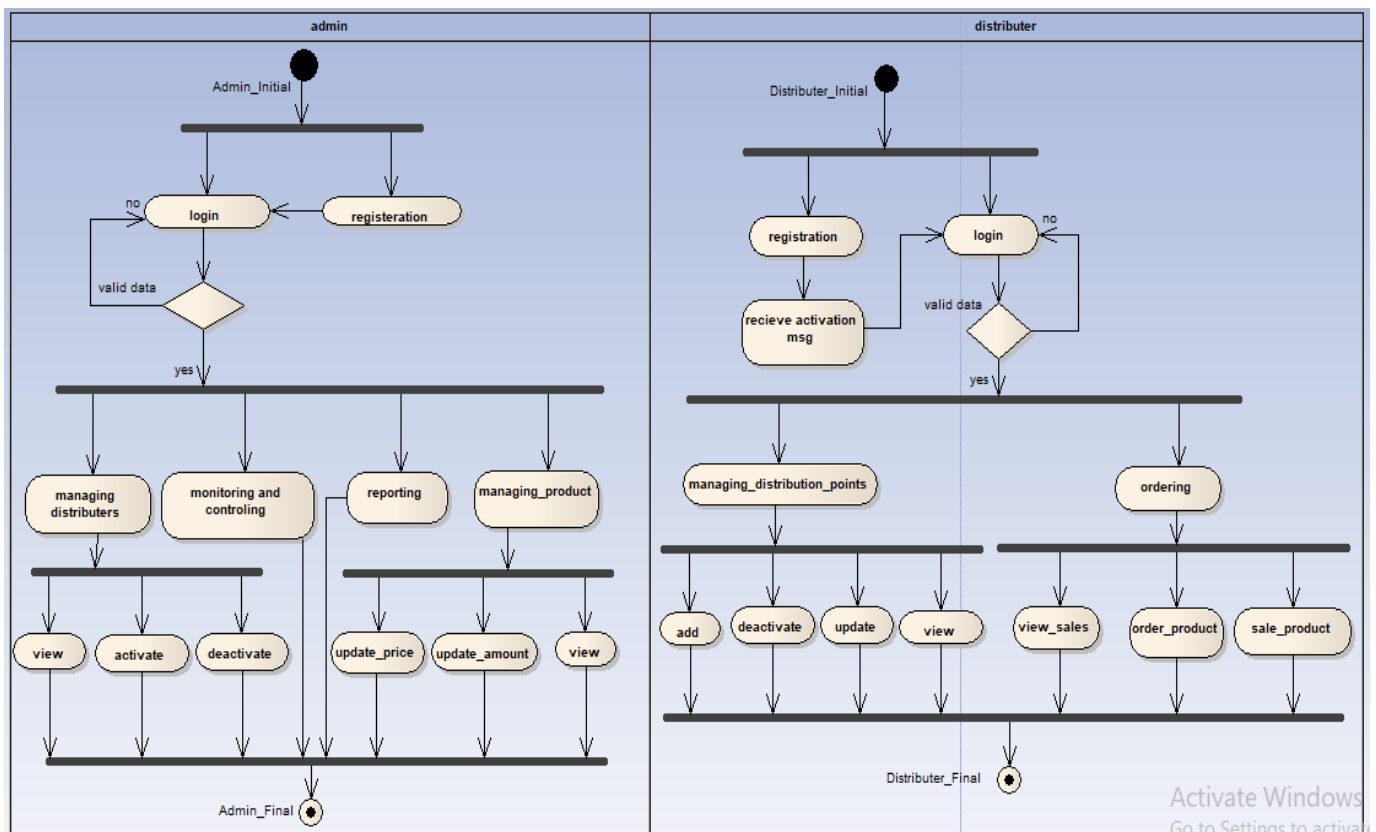


Figure 4.22 Show Activity diagram of system functions

4.5.4 Class Diagram

The figure below illustrates the class diagram of the system.

The system is consisting of 11 classes along sides their attributes, the classes are:

1. Registration.
2. Login.
3. Admin.
4. Distributer.
5. Manage_Distributer.
6. Manage_Product.
7. Manage_Points.
8. Monitoring.
9. Orders.
10. Product.
11. Sales.

The admin class extend the registration class, has a dependency with login class.

Also has relations with manage_distributer, manage_product, and monitoring classes.

The distributer class extend the registration class and the manage_distributer class, has a dependency with login class.

Also has relations with manage_points, sales, and order classes.

The sales class extends the product class.

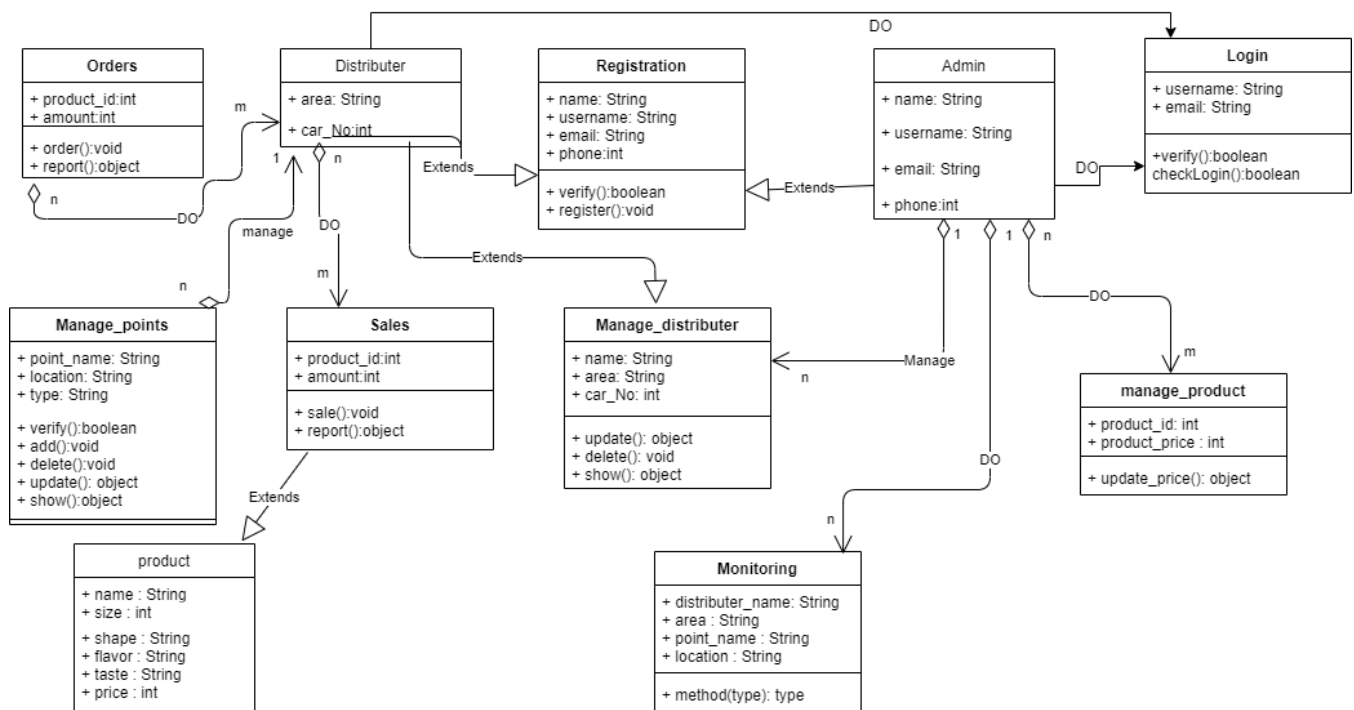


Figure 4.23 Show Class diagram of the system

4.6 Database Analysis

This section will explain the structure of the database and how it will be analyzed.

4.6.1 Entity Relationship Diagram

The figure below illustrates the entity relationship diagram which the database of the database of the system is built on it, and describes the entities of the system, their attributes, and the relationships between these entities.

The entities of this system are: admin, distributer, product, distribution point, order information, distribution point.

The admin entity has a two relations having the name mange with the distributer and product entities.

The distributer entity has a three relations, one with the distribution point named manage, distribute with the product entity, and the last is named order with the order information entity.

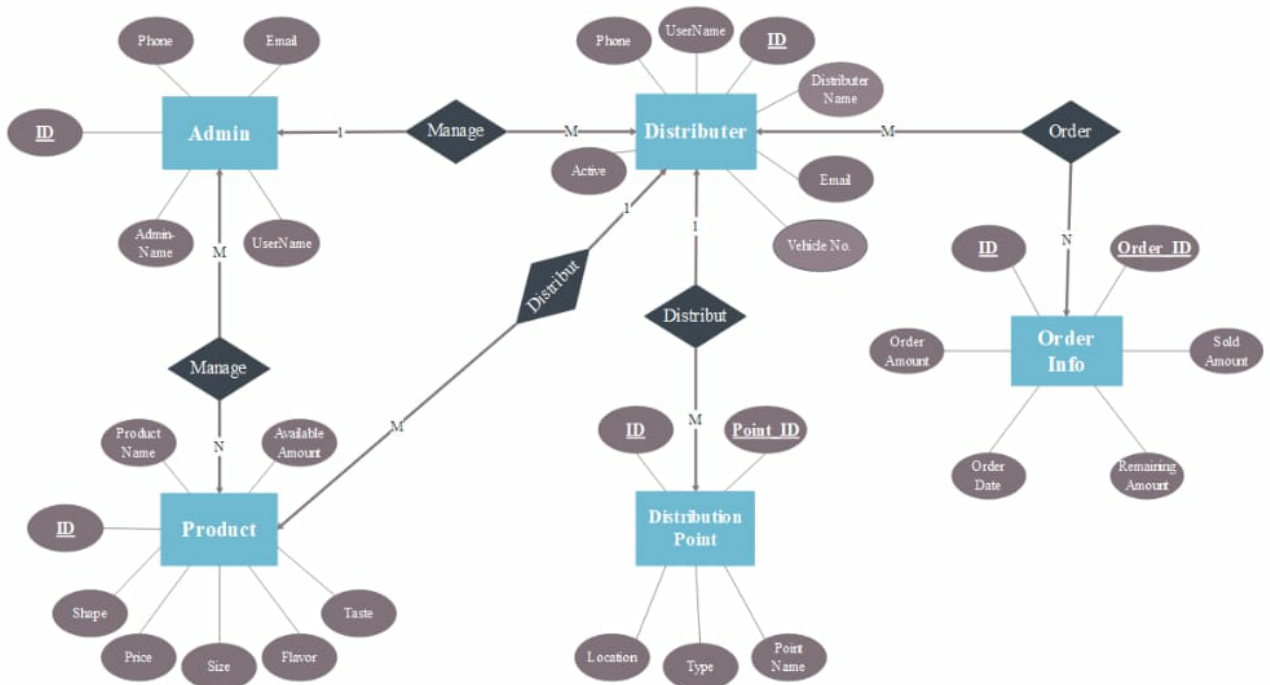


Figure 4.24 Show entity relationship diagram

4.6.2 Database Scheme description

This subsection will explain the database scheme and describe tables and their data.

Table 4.1 admin information table

FIELD NAME	TYPE	KEY
Id	Bigint(20)	Primary key PK
Name	Varchar(150)	–
Phone	Varchar(150)	–
Email	Varchar(150)	–
Username	Varchar(150)	–

Table 4.2 distributer information table

FIELD NAME	TYPE	KEY
Id	Bigint(20)	Primary key PK
Name	Varchar(150)	–
Phone	Varchar(150)	–
Email	Varchar(150)	–
Username	Varchar(150)	–
vehicle_no	Varchar(150)	–
admin_id	Bigint(20)	Unique
area_name	Varchar(150)	–
Active	Varchar(150)	–

Table 4.3 products table

FIELD NAME	TYPE	KEY
Id	Bigint(20)	Primary key PK
Name	Varchar(150)	–
Favor	Varchar(150)	–
Taste	Varchar(150)	–
Size	Varchar(150)	–
Shape	Varchar(150)	–
Price	Bigint(50)	–
available_amount	Varchar(150)	–

Table 4.4 orders table

FIELD NAME	TYPE	KEY
Id	Bigint(20)	Primary key PK
Order_id	Bigint(20)	Unique
Sold_amount	Varchar(255)	–
order_amount	Varchar(255)	–
order_date	Varchar(255)	–
distributer_id	Bigint(50)	Unique
product_id	Bigint(50)	Unique
total_price	Varchar(100)	–

Table 4.5 distribution point table

FIELD NAME	TYPE	KEY
Id	Bigint(20)	Primary key PK
Name	Varchar(255)	Unique
Phone	Varchar(255)	–
Type	Varchar(255)	–
Distributer_id	Bigint(50)	Unique
Visit	Varchar(255)	–
Latitude	Varchar(255)	–
Longitude	Varchar(100)	–

Table 4.6 distribution information table

FIELD NAME	TYPE	KEY
Id	Bigint(20)	Primary key PK
sales_id	Bigint(20)	Unique
Point_id	Bigint(20)	Unique
product_id	Bigint(20)	Unique
Distribution_date	Varchar (255)	–
Distributed_amount	Varchar(255)	–
Total_price	Bigint(50)	–

Table 4.7 sales report table

FIELD NAME	TYPE	KEY
sales_id	Bigint(20)	Primary key PK
point_id	Bigint(20)	Unique
distributer_id	Bigint(20)	Unique
Total_price	Bigint(20)	–

Table 4.8 orders reports table

FIELD NAME	TYPE	KEY
order_id	Bigint(20)	Primary key PK
distributer _id	Bigint(20)	Unique
Total_price	Bigint(20)	–

Table 4.9 product management table

FIELD NAME	TYPE	KEY
admin_id	Bigint(20)	Primary key PK
product _id	Bigint(20)	Unique

Table 4.10 show sales report view

FIELD NAME	TYPE	KEY
Sales_id	Bigint(20)	Primary key PK
point_id	Bigint(20)	Unique
Distributer_id	Bigint(20)	Unique
Total_price	Bigint(20)	–
D_name	Varchar(255)	–

4.7 Summary

This chapter deals with the aspects of analyzing and designing system requirements using UML, in addition to analyzing and designing the system database using ERD based on the requirements collected, and explaining the system architecture. The next chapter will explain how these requirements, which were analyzed and designed were implemented as a software system.

CHAPTER FIVE

IMPLEMENTATION

5 Introduction

This chapter explains the application process and the functions that the system performs, whether at the level of a web application or a mobile application.

5.2 Web Application

5.2.1 Registration Screen

This is the registration process in the web application, the admin enters his name, phone number, email and username.

The image shows a registration screen for a web application. On the left, there is a vertical purple-to-blue gradient bar. To its right, a white rounded rectangle contains the registration form. At the top of this rectangle is a logo of a truck with a gear, labeled 'Distributor'. Below the logo is the title 'انشاء حساب جديد' (Create New Account) in blue. The form consists of five input fields with Arabic labels: 'اسم الادمين' (Admin Name), 'رقم الهاتف' (Phone Number), 'البريد الالكتروني' (Email), 'اسم المستخدم' (Username), and a final 'انشاء حساب' (Create Account) button with a blue-to-purple gradient. Below the button is a link 'لديك حساب مسبقا؟ تسجيل دخول' (Already have an account? Log In).

Figure 5.1 show registration process in web application

5.2.2 Login Screen

This is the process of logging in to the web application, and it is done when the admin enters his username and email to be able to access the system to perform his tasks.

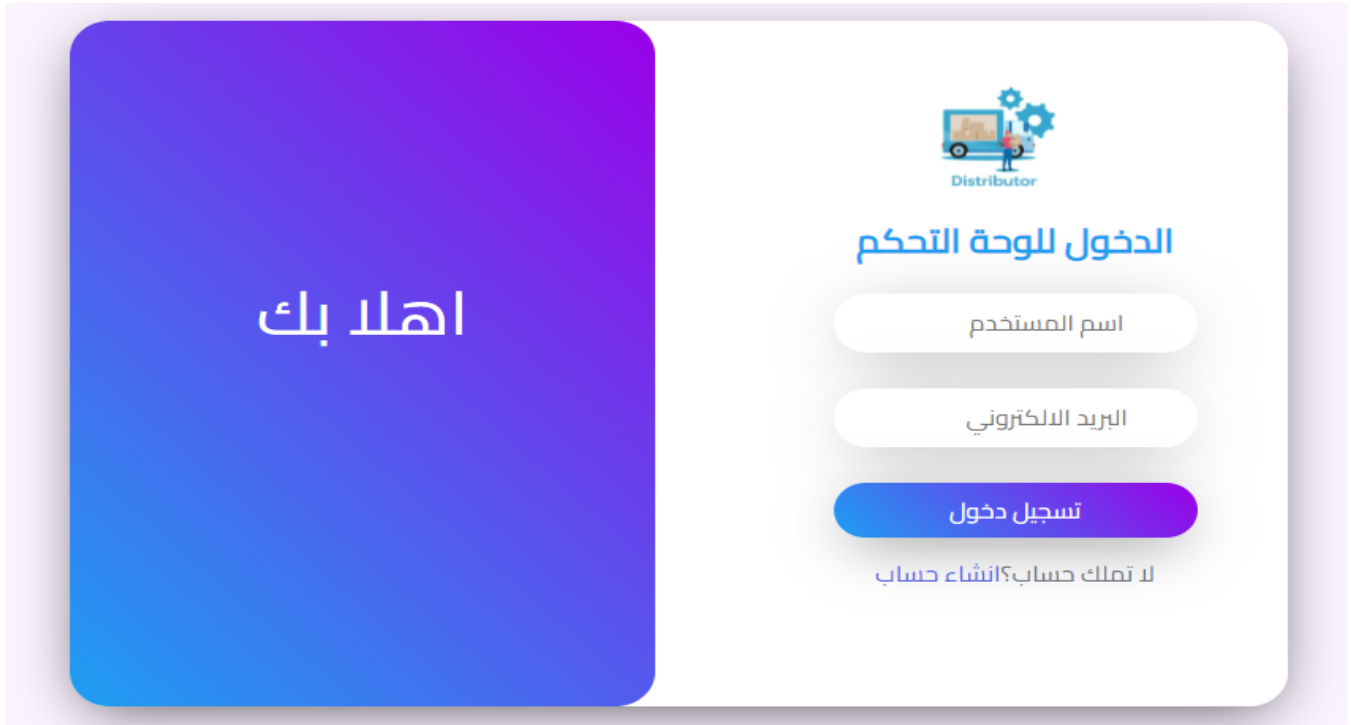


Figure 5.2 show login process in web application

5.2.3 Dashboard Screen

This is the dashboard for the web application that shows the total sales of each product, the product orders table, the top five sales table and the total of the top five orders.



Figure 5.3 show dashboard in web application

5.2.4 Distributers Management Screen

This is the process of managing distributors, where the admin can search for distributor, view the points of sale for which the distributor is responsible, view a map of his points of sale, and deactivate the distributor's account.

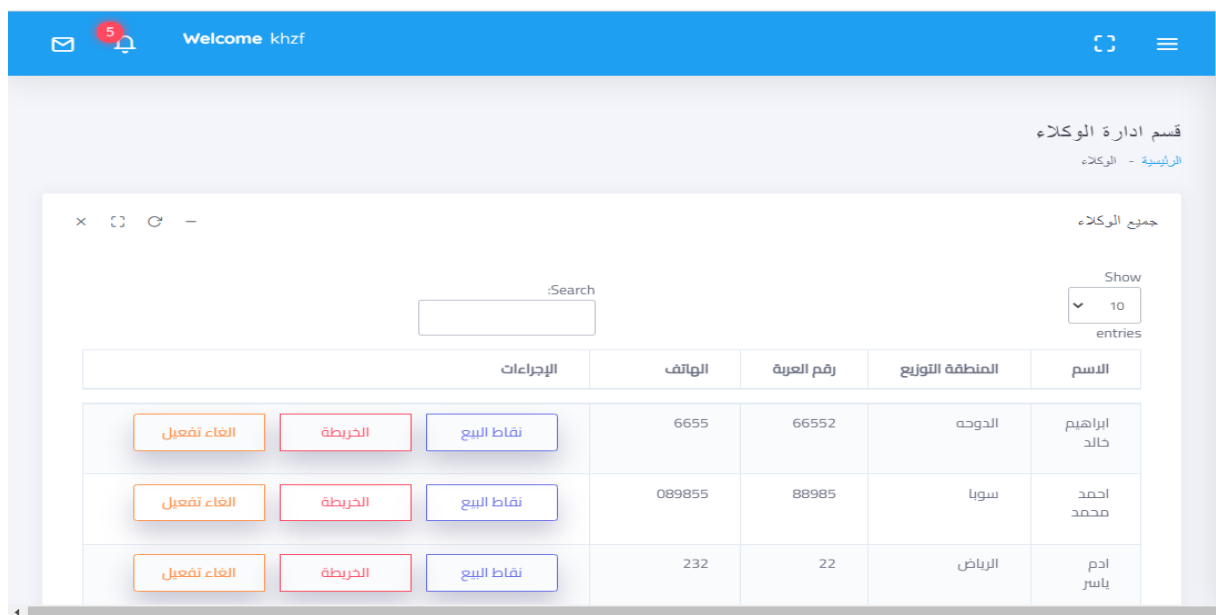


Figure 5.4 show distributors management process in web application

5.2.5 Point of Sale Management Screen

This is the point of sale management process, where the admin can view the points of sale of each distributor, and display all the distributors' point of sale on the map.

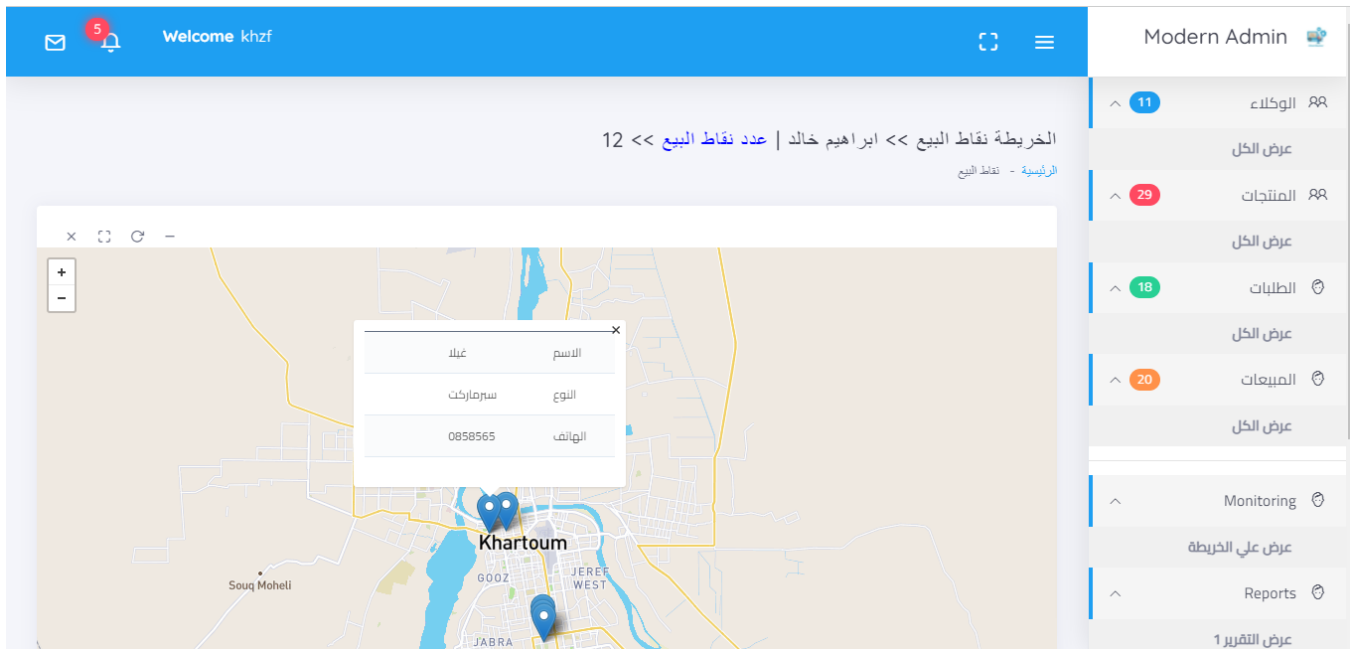


Figure 5.5 show point of sale management process in web application

5.2.6 Products Management Screen

This is the process of product management, where the admin can search for a specific product, view all products, details of each product and the amount available for each product, and also can update the product price.

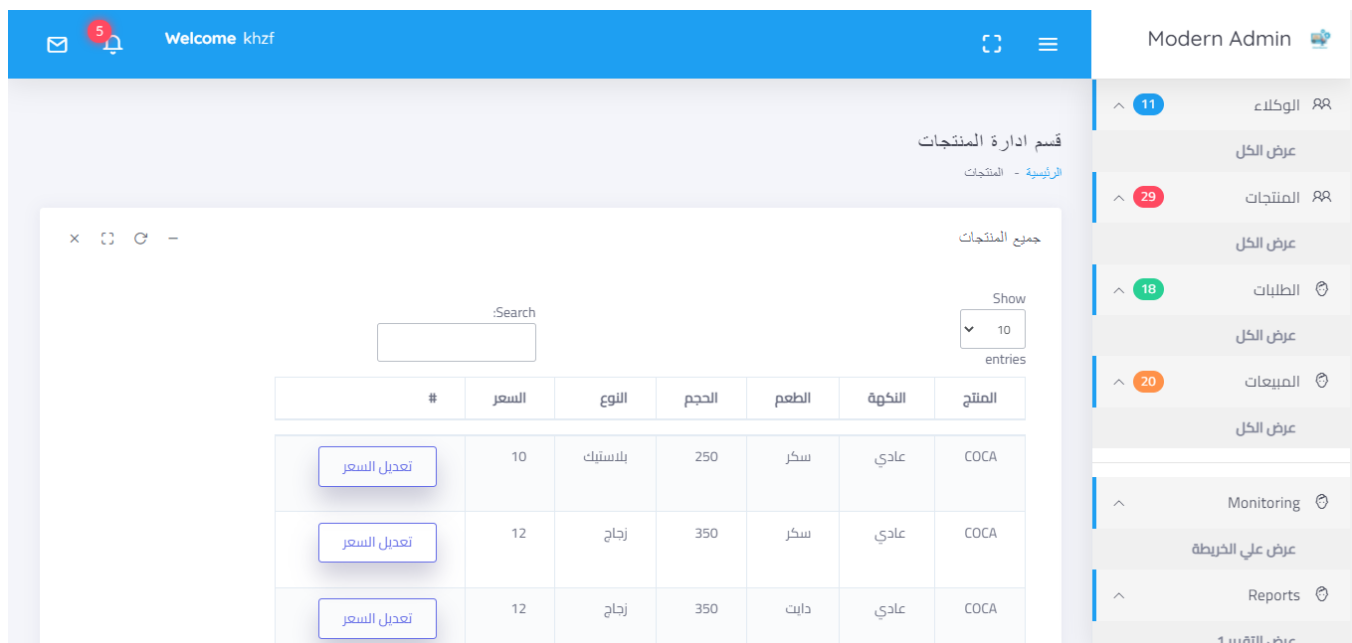


Figure 5.6 show products management process in web application

5.2.7 Orders Management Screen

This is the process of managing orders, where the admin can search for a request, and know the request information such as the distributor who made the order, and the total and date of the orders.

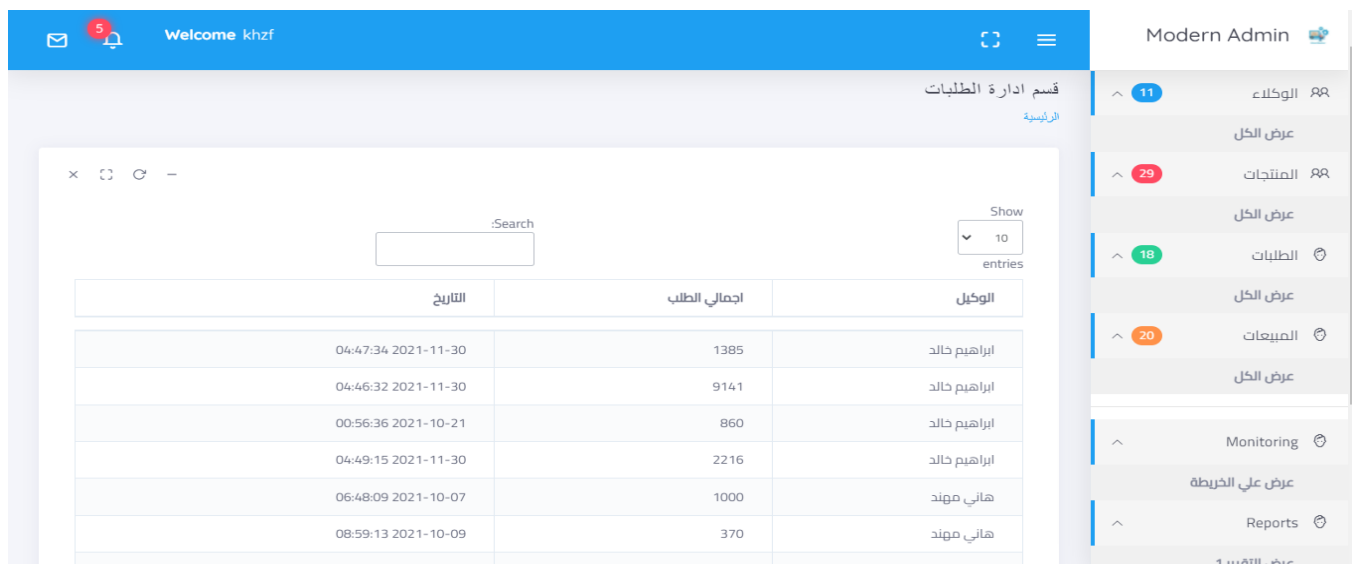


Figure 5.7 show orders management process

5.2.8 Sales Management Screen

This is the sales management process, where the admin can view the sales of each order made by a particular distributor, and

search for sales made by a particular distributor.

التاريخ	اجمالي الفاتورة	الوكيل
04:47:34 2021-11-30	1385	ابراهيم خالد
04:46:32 2021-11-30	9141	ابراهيم خالد
00:56:36 2021-10-21	860	ابراهيم خالد
04:49:15 2021-11-30	2216	ابراهيم خالد
06:48:09 2021-10-07	1000	هاني مهدي
08:59:13 2021-10-09	370	هاني مهدي

Figure 5. 8 show sales management process

5.2.9 Monitoring and Tracking Process Screen

This is the monitoring and tracking process which is the core feature of this system, where the admin can monitor and track each distributor and their distribution process on a map and in real time.

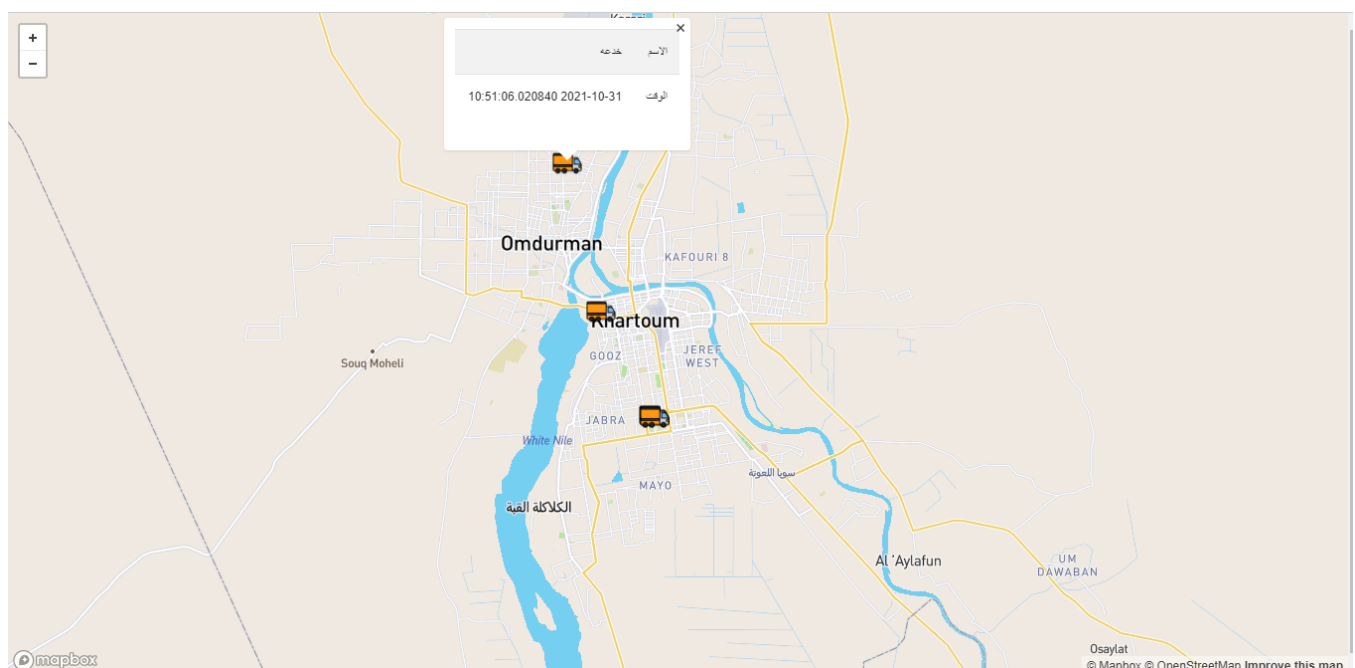


Figure 5.9 show Monitoring and Tracking process

5.2.10 Reporting process Screen

This is the reporting process, where reports are presented by visualizing distribution data as heat maps, clustered maps, and charts, showing important information about distribution such as the best-selling product in a particular distribution area, and the total sales for each product in each distribution area.

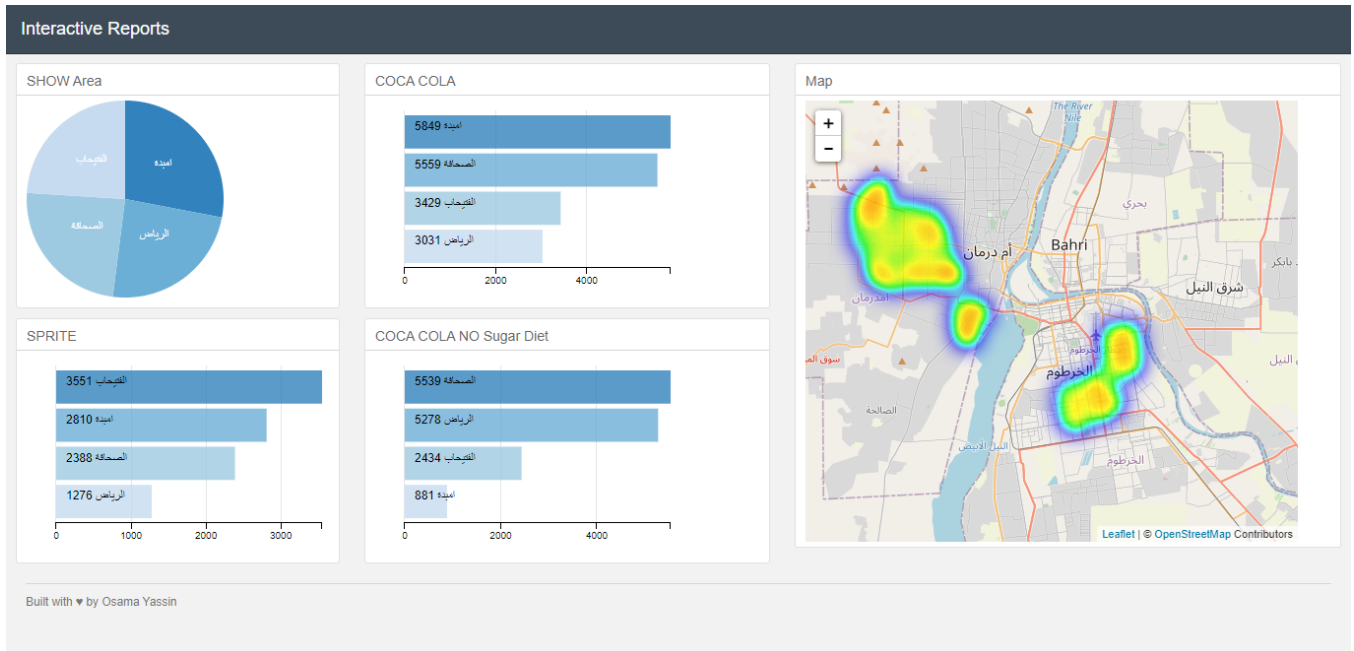


Figure 5.10 show Data visualization as chart and clustered maps

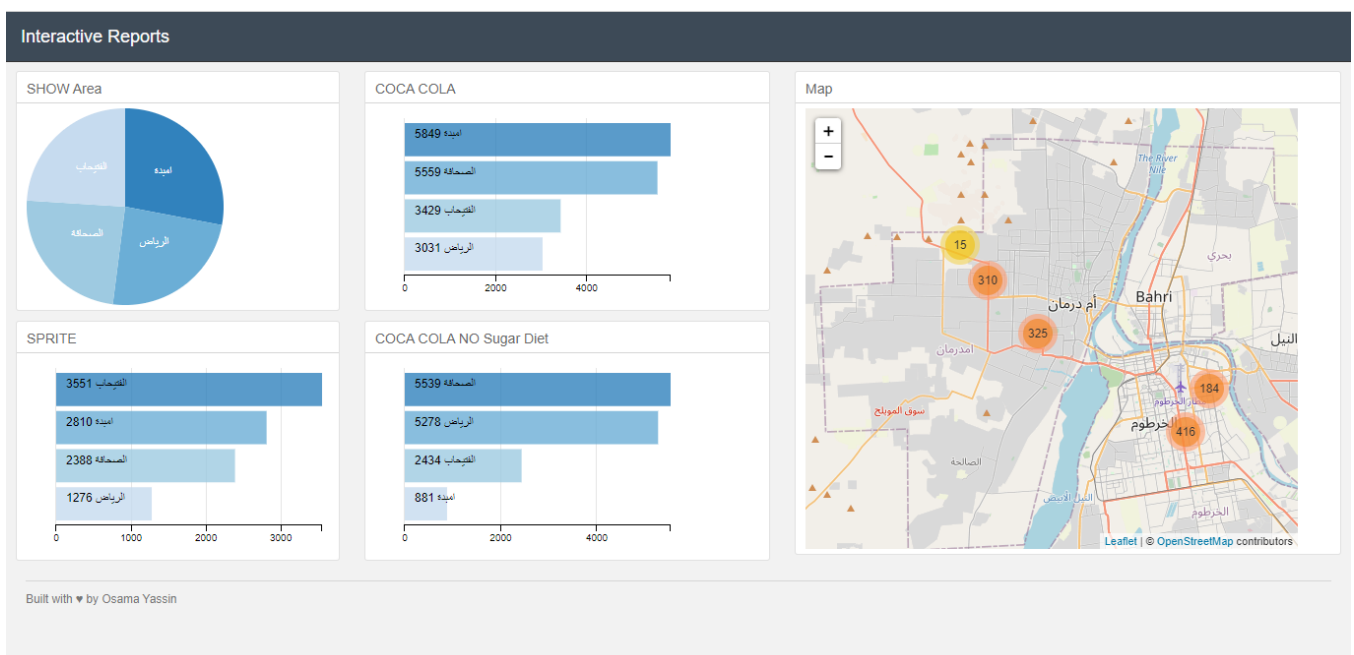


Figure 5.11 show Data visualization as chart and heat maps

5.3 Mobile Application

5.3.1 Registration Screen

This is the registration process in the mobile application, the distributor must enter his full name, username, email, phone number, vehicle's number and distribution area.

مرحباً بك
اشترك للمتابعة

التسجيل
تسجيل الدخول

اسم كامل

اسم المستخدم

البريد الإلكتروني

رقم الهاتف

رقم العربية

منطقة التوزيع

Figure 5.12 show registration process in mobile application

5.3.2 Login Screen

This is the process of log in to the mobile application, and it is done when the distributor enters his username and email to be able to access the system to perform his tasks.

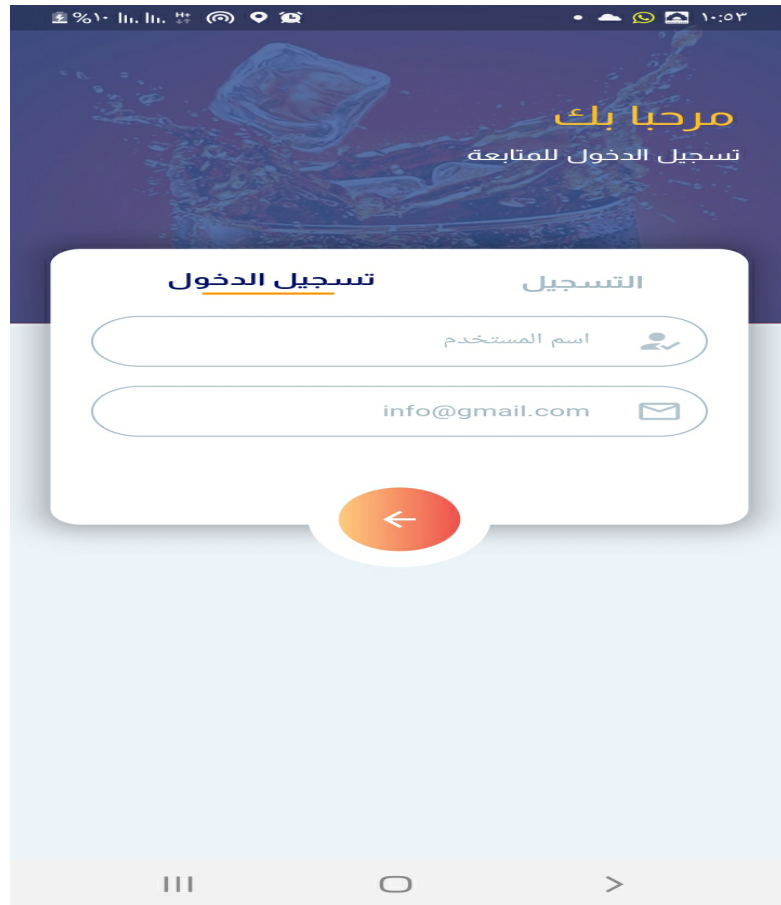


Figure 5.13 show login process in mobile application

5.3.3 Dashboard Screen

This is the dashboard for the mobile application that shows the total number of orders the distributor has made, total number of distribution points he is responsible to cover, and number of distribution points he already covered. He is also able to reset that information after covering all distribution points in his area.



Figure 5.14 show dashboard in mobile application

5.3.4 Menu Screen

This is the screen from where the distributor can choose the task he wants to perform by clicking on that task.

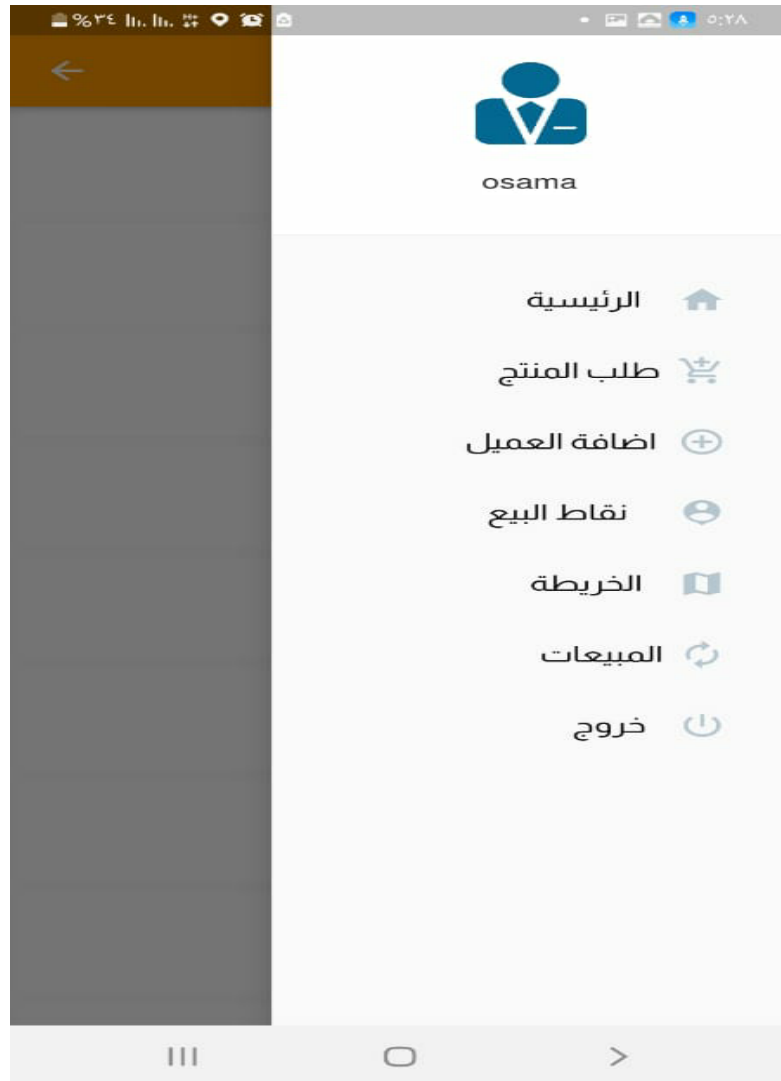


Figure 5.15 show menu in mobile application

5.3.5 Order products process Screen

This is the order products process the distributor must be able to request the specified products with specified amount that he needs to distribute in a particular distribution points.

طلب المنتجات

كوكاكولا - سكر

000 350 - زجاج

000 250 - قوارير

000 400 - قوارير

000 500 - قوارير

000 قوارير 1 لتر

كوكاكولا - داييت

000 350 - زجاج

000 250 - قوارير

000 400 - قوارير

000 500 - قوارير

000 قوارير 1 لتر

Figure 5.16 show order product process

5.3.6 Add new distribution process Screen

This is the process of adding new distribution points discovered by distributor in his distribution path, which the distributor needs to enter the name of distribution point, the phone number of distribution point's owner, and the type of distribution point.

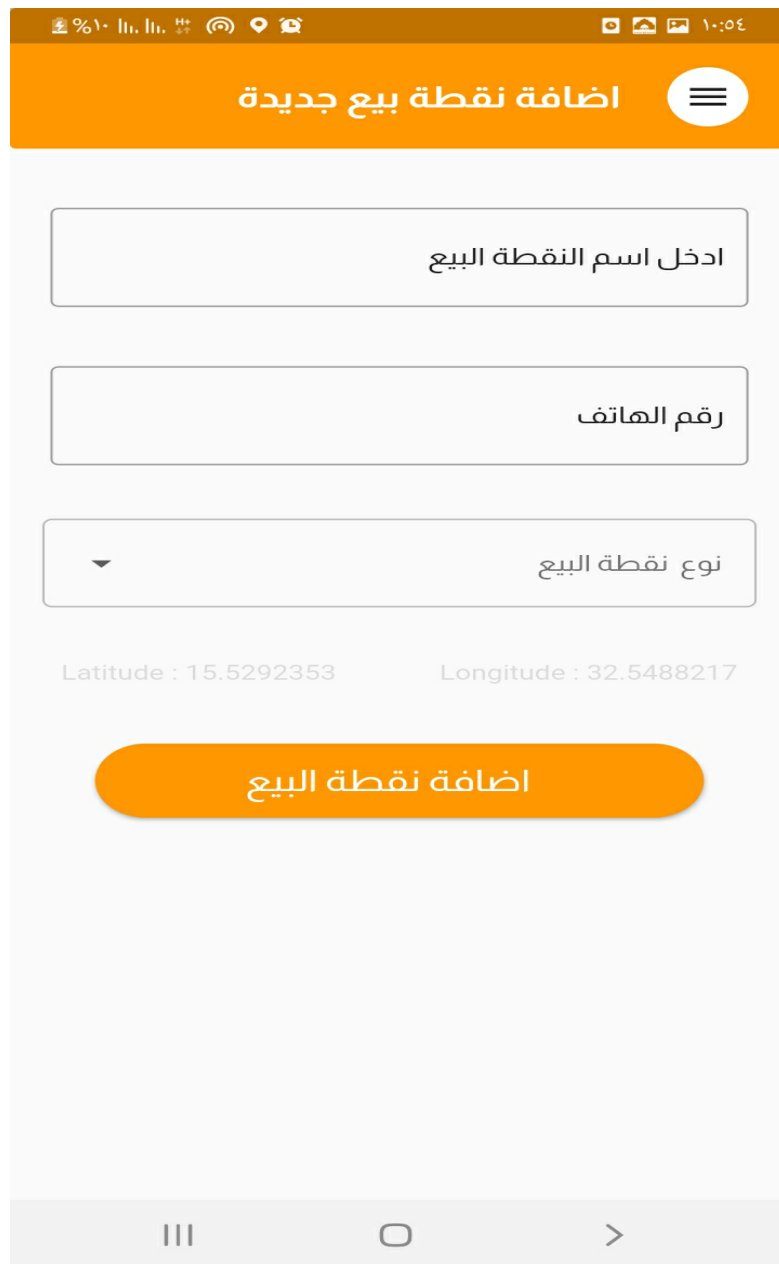


Figure 5.17 show add new distribution process

5.3.7 Distribution points management process Screen

This is the process that lists the distribution points associated with the operations that the distributor can perform to manage distribution points which are deleting, and display location of a specific distribution point.

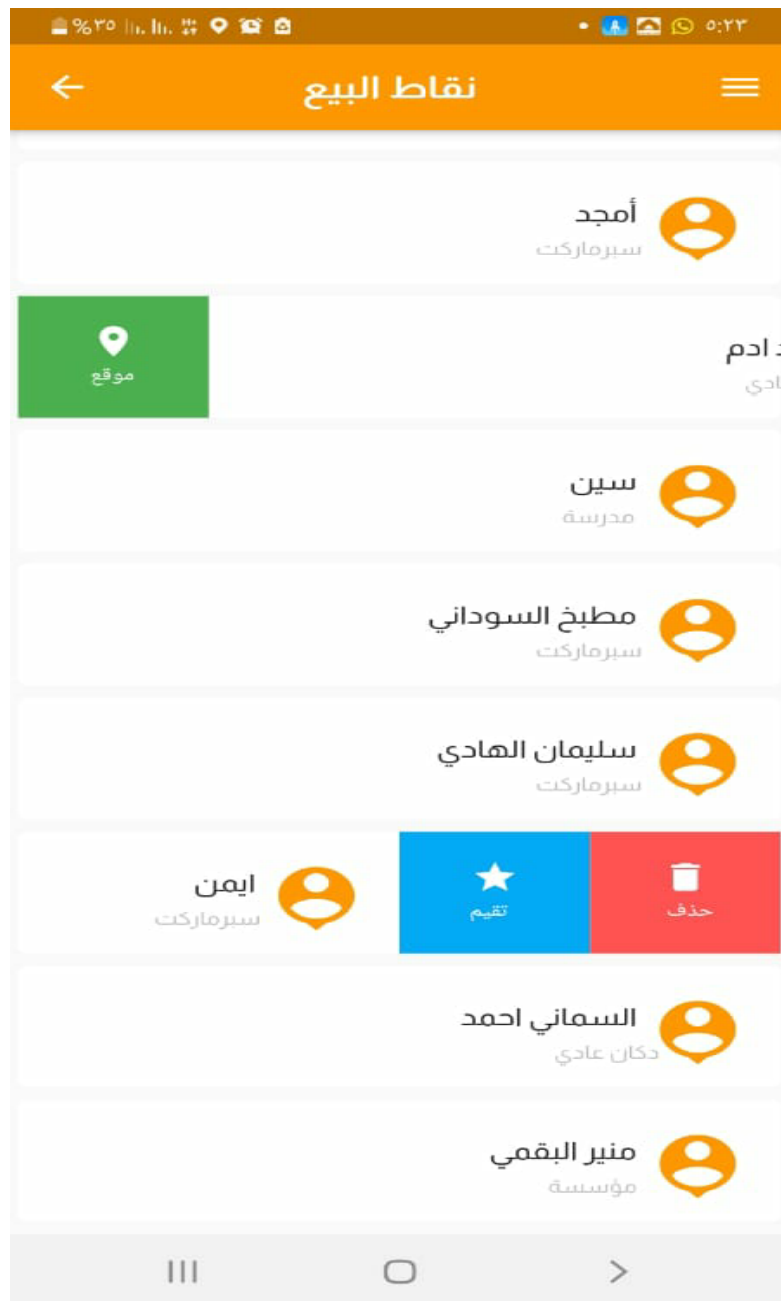


Figure 5.18 show distribution points management process

5.3.8 Display specific distribution point location Screen

This is the screen that shows the location of a specific distribution point along with its name on the map.



Figure 5.19 show the display of a specific distribution point location

5.3.9 Selling products process Screen

This is the process of selling products to specific distribution point in which the distributor can only perform a sale operation when he is in the range of the distribution point (200-meter radius).

اسم: مهند فتحي
النوع: كشك

قوارير - 500	000
قوارير - 1 لتر	000

فانتا - فراولة

زجاج - 350	000
قوارير - 250	000
قوارير - 400	000
قوارير - 500	000
قوارير - 1 لتر	000

بيع المنتجات

Figure 5.20 show the selling process

5.3.10 Display all distribution points locations Screen

This is the screen that shows the locations of all distribution points belongs to a specific distributor along with their names on the map.



Figure 5.21 show the display of all distribution points locations

5.3.11 Total sales quantities Screen

This is the screen that shows the total quantities of a products sold by the distributor to distribution points represented by sales price.



Figure 5.22 show total sales quantities represented by sales price

5.3.12 Logout Screen

This is the screen that shows the operation of logging out of the application.

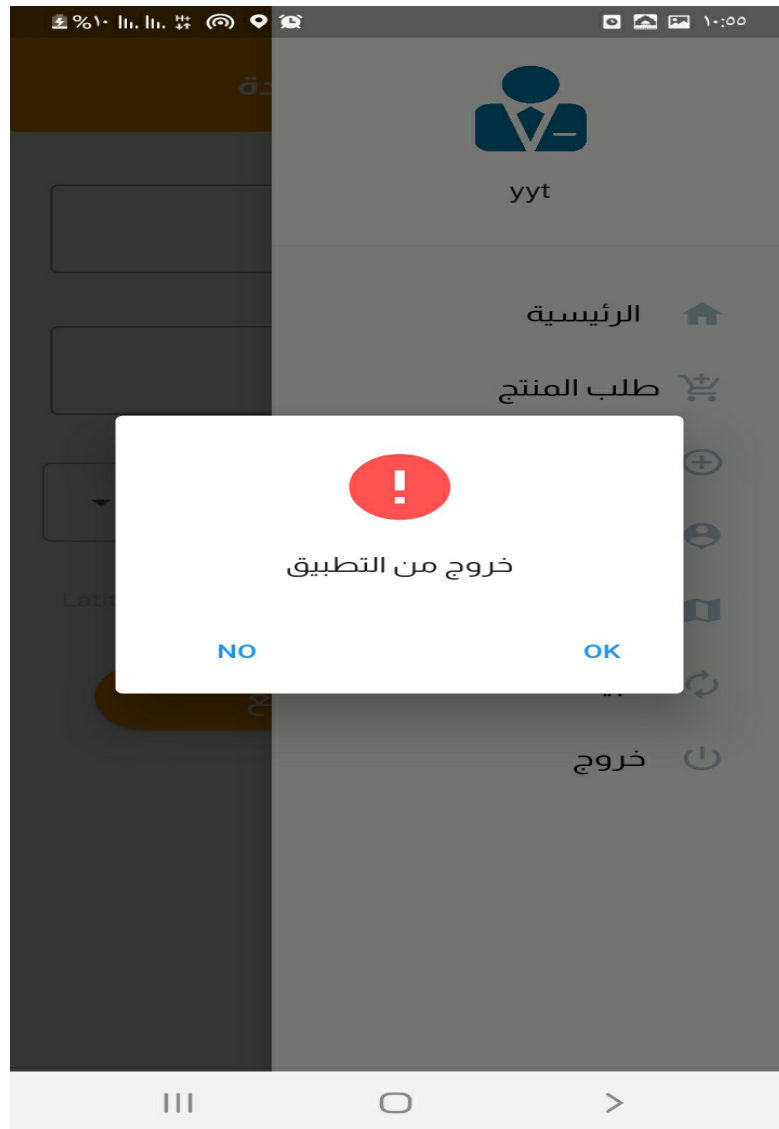


Figure 5.23 show the logout of the application

5.4 Summary

This chapter illustrate the implementation of the web and mobile applications shown in screens and how the overall system works.

The next chapter will discuss the conclusion that came up after doing this research and recommendations.

CHAPTER SIX

CONCLUSION

&

RECOMMENDATIONS

3 Introduction

This chapter discusses the results have been achieved after System is implemented, recommendations for future researches and conclusion.

6.2 Results

After this research was completed the following results were achieved:

1. A smart logistic services system was developed. through which the process of distributing products and goods is done in an optimal way and imposing control over the distribution process.
2. The cost and time required in the distribution process were reduced and the resources of the distribution companies had been optimized.
3. The profits of products were increased.
4. Time and effort had been reduced for the consumer by knowing the exact time of arrival of the product to the place of consumption.
5. The system provides the feature of adding new distribution (consumption) areas and approving them thus covering the market had been achieved.

6.3 Recommendations

1. Adding the billing feature after doing a sale operation.
2. Expand the system's scope to cover more than just soft drinks products.

6.4 Conclusion

After completing this research, we concluded that the management of product distribution and monitoring of the transportation process by means of a software system led to solving the problems experienced by the traditional method, and this procedure also helped in making administrative decisions related to marketing products in distribution areas in an improved manner based on the Reports generated by the system.

References

- [1] Logistic services <https://redstagfulfillment.com/what-are-logistics-services/>
- [2] Researching the effect of logistics on the competitive advantages of container transport in Port Sudan port [حامد، أثر اللوجستيات فى الميزة التنافسية للنقل بالحاويات فى، مريم فرج محمد 4-8-2015م، 2014م الى 1995 ميناء بورتسودان].
- [3] Ph.D Research on the effect of applying the logistical management approach on the performance of industrial production institutions in Sudan [لامام، محمد عبدالرحمن مصطفى] اثر تطبيق منهج الادارة اللوجستية فى أداء مؤسسات الإنتاج الصناعي فى السودان 2009-2000 1-5-2010]
- [4] The use of information technology in logistical activities and its impact on achieving a competitive advantage for the organization <http://dspace.univbatna.dz/xmlui/handle/123456789/3> [2018].
- [5] Halal Economics Opportunities and Challenges <https://ssrn.com/abstrac9> [February 12,2018].
- [6] Integrating BIM and GIS to improve the visual monitoring of construction supply chain management CSCM <https://www.sciencedirect.com/science/article/abs/pii/S0926580512002440> [August 27 ,2009].
- [7] Unified Modeling language (UML)

<https://www.gliffy.com/blog/what-is-uml-everything-you-need-to-know-about-unified-modeling-language#common-notations> .

[8] use case diagram

<https://whatis.techtarget.com/definition/use-case-diagram> .

[9] sequence diagram

<https://www.lucidchart.com/pages/uml-sequence-diagram> .

[10] activity diagram

<https://www.lucidchart.com/pages/uml-activity-diagram> .

[11] class diagram <https://www.lucidchart.com/pages/uml-class-diagram> .

[12] Entity Relationship Diagram ERD

<https://www.lucidchart.com/pages/er-diagrams> .

[13] HTML on <http://www.w3schools.com/hml/> .

[14] CSS on <http://www.w3schools.com/css/> .

[15] Bootstrap on <https://getbootstrap.com/> .

[16] Laravel - PHP framework on

<https://laravel.com/docs/8.x> .

[17] Visual Studio Code on <https://code.visualstudio.com/> .

[18] Flutter framework on

<https://www.freecodecamp.org/news/what-is-flutter-and-why-you-should-learn-it-in-2020/> .

[19] android studio on

<https://www.techtarget.com/searchmobilecomputing/definition/Android-Studio> .

[20] ArcGIS on <https://www.gistandards.eu/what-is-arcgis/> .